Simpler to use, rich in information

Revision 4 - simple construction navigation with company, currency, reporting and AI requirements.

Simple navigation needs a strong information foundation. Keep project work in one continuous workspace, with company-wide specialist queues opening the same records. This revision adds explicit company ownership, vendor currencies, scoped reporting, report findings and carefully placed AI assistance.

The revised recommendation

Starting point	Purpose
My Work	What needs my action or decision, what I am waiting for, and where I resume.
Tenders	Opportunity, tender, estimate, offer and award, with a clear project handoff.
Projects	One workspace: Overview, Plan, Site, Buying, Costs & Contracts, Documents, Handover.
Company	Visible directory for Buying, Finance, People, Assets & Fleet, Stores, Contacts & Catalogue, and Reports.

Keep project, location, work item, owner and next action visible. A case continues through its stages; users do not repeatedly open the same tabs in successive modules. Detailed quantities, money, evidence and history remain available under clear labels.

What the study covers

Mapped **all 42 current primary entries** and **all twelve requested capabilities**; walked twelve findability scenarios on paper; examined four difficult cross-team journeys; specified record behaviour, scope, search, permissions and validation gates. No staff usability test or executable prototype was used.

Read this first

Pages 2-5: the recommended map and daily experience.
Page 6: exact homes for your twelve areas.
Pages 7-8: tender-to-handover and difficult journeys.
Pages 9-11: every current primary menu mapped.
Pages 12-13: findability checks and user-testing criteria.

Supporting study: existing UI findings and screenshots, pages 14-22; capability existence/need and detail, pages 23-38; original-code scope and validation, pages 39-41; evidence and limits, pages 42-43. Pages 2-13 replace the previous candidate navigation proposal.

New requirements, pages 44-53: multiple companies; vendor-based purchase currencies and an FX example; project/company/group reporting; summaries driven by report findings; AI placement, controls, rollout and acceptance checks.

Status: this revision extends the study, not the live application. Earlier observations remain dated 28 October 2026 and are not newly retested. It is a recommended information map, not proof of the best possible usability. No application source was copied or new ERP code built.

Four clear starting points

Recommended information map. This is a study proposal, not a tested interface or a visual design.

Start with a task, a tender, a project, or company work. Keep these four destinations in a small global navigation. After opening a project, use its seven stable areas below. Do not keep the old module tree alongside them.

Start here	What the user expects to find
My Work	What I need to do, decide, chase or resume. Direct links to my frequent specialist queues and personal requests.
Tenders	Opportunities, bid/no-bid, tender documents, estimates, offers, revisions and award. A tender remains traceable after a project is created.
Projects	Find a project, then manage its plan, site work, buying, costs, documents and handover in one context.
Company	Buying across projects, Finance, People, Assets & Fleet, Stores, Contacts & Catalogue, and Reports. Show these names visibly on the directory.

The project map

Project areas - stable order	Plain-language purpose
Overview | Plan | Site	Understand the project; plan the work; record and control what happens on site.
Buying | Costs & Contracts	Obtain materials/subcontract work; explain budgets, obligations, changes, claims and money.
Documents | Handover	Find controlled information; prove work is ready, accepted and supported after completion.

Utilities, not additional business modules

Search, company/project scope, help, profile and authorised Administration access stay consistent. The Company workspace follows the selected legal entity; authorised group views explicitly show their included entities. A project overview opens on current work, risks and next actions.

Why this replaces Revision 2: Buying, Finance and Resources no longer compete with Projects in the primary menu. Company-wide specialists use direct queue shortcuts; project users follow the job without leaving its context. The number four is a proposed grouping, not a usability result.

Seven places inside a project

Each area opens a useful work view with clearly named links. No nested accordion tree and no universal row of repeated tabs.

Area	Visible contents and boundaries
Overview	Next actions, blockers, due milestones and a small role-relevant performance summary. Setup checklist and project facts are secondary links. KPIs link to source records.
Plan	Programme; Look-ahead; Constraints & delays; Resources. Baseline/current dates and progress stay connected. Delay events link to commercial/time claims, without copying them.
Site	Daily work; Materials; Quality & NCRs; Safety; Waste. Direct actions: Record daily work, Request materials, Receive delivery, Report a quality problem. Show assigned field work first.
Buying	Requests; Orders & deliveries; Subcontract packages. Follow selection/award, delivery or measured work through the same order/package. Company Buying is an all-project view.
Costs & Contracts	Budget & cost; Client contract; Changes; Claims & payments; Cash flow; Final account. Subcontract amounts link to their Buying package. Finance posting/bank work stays in Company Finance.
Documents	Drawings; Submittals; RFIs; Specifications & methods; Correspondence. Current/revision/issue/review states are explicit. Source-case attachments are searchable links, not copied documents.
Handover	Snags; Testing & commissioning; Handover packs; Defects & warranties. Organise by location/system/installed asset, with partial handover and client acceptance evidence.

Stable structure; relevant content

Keep area names and order stable across lifecycle stages. Stage and role change the recommended work and visible permissions, not the meaning or location of an area. Snagging and handover preparation remain reachable during construction.

A specialist registry can be a dense, filterable table when that helps its user. A site worker gets a focused action and the evidence needed now. Neither requires a second copy of the data. Area sublinks are page content, not another permanently expanded sidebar. Preserve filters and scroll position when returning from a record.

My Work is the daily starting point

Role-aware priorities, with consistent navigation and explicit authority. Proposed defaults must be checked with actual staff.

Show **Needs my decision**, **Assigned to me**, and **Waiting on others** as views of one queue. Every row says what to do, project/location, due date, current owner and why it is waiting. Resume drafts and recent work directly; routine notifications must not bury decisions.

Person	Default work and direct shortcuts
Owner / project manager	Exceptions, milestones, cost/cash outlook and delegated decisions. Project Overview and portfolio reports.
Estimator / tender manager	Bid deadlines, estimate review and offer revisions. Tender workspace.
Site engineer / supervisor	Today's work, drawings, material needs, inspections and blockers. Site plus the look-ahead.
QS / cost controller / buyer	Measurements, changes, forecasts, claims and purchasing decisions. Budget & cost or a cross-project Buying queue.
QA/QC / document controller	Inspections, NCR actions and document reviews; overdue owners visible. Quality & NCRs or Documents.
Finance / people / plant teams	Authorised payment/close work, staff requests, or asset service/availability. Their named Company workspace directly.

Company is a visible directory, not a miscellaneous menu

Buying - all-project procurement and subcontract queues.
Finance - receivables, payables, cash/bank, ledger, period close and financial reports.
People - employment, recruitment, leave, attendance and time.
Assets & Fleet - custody, hire, maintenance and vehicles.
Stores - warehouses, transfers, counts and stock movements.
Contacts & Catalogue - parties, materials/services and shared units.
Reports - cross-project comparisons and governed metric/report definitions.

Daily specialists go straight to their queue from My Work. Keep company and All projects / named project scope visible; a cross-company queue labels each record. Roles and entity permissions control access. Company > Reports uses the same definitions as project reports, always with a findings summary.

One case, a clear next step

Screen behaviour requirements. The proposal preserves rich information while reducing repeated navigation and data entry.

Layer	What stays visible or opens on demand
Always visible	Record title/ID; project and relevant package/location; status in words; owner; next action and due date; material blockers; version/cutoff for decisions and totals.
Main working content	The information needed for the current task and one primary action, with explicit secondary alternatives such as Return for correction. A reviewer sees evidence before deciding.
Named detail sections	Quantities, resources, money, technical evidence or terms as appropriate to the record. Show section names and key warnings; expand detail when useful. No vague More container for essential work.
Related records and history	Linked order, certificate, bill, inspection, change or drawing; decision and revision history. Open a related record with context retained and a clear return path.

A record evolves; the user does not restart

An NCR stays the same case through containment, remedy, verification and closure. A subcontract package keeps scope, work and money together; its certificates and bills remain separately controlled linked records. Do not copy every tab into each linked screen or flatten legally different transactions into one form.

Prefill known project, item, location and party details. Ask only for facts needed at the current transition. Preserve drafts, validation messages and entered information. New approved versions require explicit revision; historic decisions retain the evidence actually reviewed.

Contextual AI can help find, explain or draft information without becoming another navigation layer. Keep its sources and active company/project visible. Financial arithmetic and approval rules remain controlled logic; AI proposals require the relevant review. See pages 50-52.

Find information without knowing its module

Search by ID, title, party, floor/unit, asset tag and familiar aliases: NCR / non-conformance / quality problem; VO / variation / change; plant / equipment; snag / defect. Results show type, scope, state and current revision, deduplicated by record identity. Always offer a browsable route as well.

Make search scope obvious and offer a deliberate broader search. Respect permissions in results, snippets, exports and links. Distinguish no records, active filters, insufficient access, stale data and a loading/error state. Attachment full-text/OCR coverage must be stated and verified.

Exact homes for all twelve areas

All project paths below start inside the selected project unless Company or My Work is specified. These are proposed homes, not live routes.

Capability	Home	Owner / connected work
NCR operations	Site > Quality & NCRs	QA/site: contain, investigate, authorise remedy, verify. Linked cost, waste, change and delay evidence.
Asset management	Company > Assets & Fleet	Plant team: availability, custody, hire and maintenance. Project assignment/use; installed assets in Handover.
Payments & cash flow	Costs & Contracts > Claims & payments / Cash flow	QS/finance: project outlook and collections. Company > Finance handles bank, payment and reconciliation work.
Subcontracting	Buying > Subcontract packages	Procurement/QS: scope, award, work, changes and money stay on one package; site reviews assigned work.
Item cost analysis	Costs & Contracts > Budget & cost	Cost controller: choose item/location, then compare budget, incurred, cash spent and forecast with source drill-down.
Waste management	Site > Waste	Site/stores: material loss, reuse and disposal evidence. Link stock movement, carrier/destination and allocated cost.
Variations / changes	Costs & Contracts > Changes	QS/PM: one change case from instruction through cost/time decision. Site and Documents provide direct Raise change links.
Programme & delay	Plan > Programme / Constraints & delays	Planner: baseline/current dates, progress, constraints and delay cases. Link an EOT assessment to the same events.
KPI management	Overview > Performance	PM: relevant exceptions and trends with owner, cutoff and definition. Company > Reports maintains shared definitions.
Keeping records	Documents + each source record	Document controller owns controlled revisions; evidence stays attached to its transaction/case and is searchable centrally.
Approval workflow	My Work + record decision area	Approver: see evidence, version, authority, next owner and deadline. Policies live in Administration.
Tracking / transparency	Every record; Overview exceptions	All permitted roles: state, blocker, responsible person, next action and history; totals lead to source facts.

Records, approval decisions and tracking are present on every relevant case. Their dedicated search or queue views are additional ways to reach the same information, not additional copies or independent workflows.

The construction journey stays connected

A project lifecycle is a guide, not a compulsory wizard. Procurement, design, site work, commercial work and handover preparation overlap.

Stage	Where work starts	Handoff that must be explicit
1. Opportunity / bid decision	Tenders > Opportunity	Client, tender requirements, deadlines, scope, risks and accountable bid decision.
2. Estimate / offer	Tender workspace	Take-off/BOQ, resource rates, quotes, exclusions and approved offer revision. Preserve the tender basis.
3. Award / mobilisation	Award -> Project setup checklist	Accepted contract and scope; versioned cost baseline; location/work-item structure; programme; authorities and obligations. Direct awards have a supported entry.
4. Design / plan / procure	Documents + Plan + Buying	Approved information, material/subcontract readiness and site prerequisites linked to planned work.
5. Execute / inspect	Site, from the look-ahead	Daily evidence, quantities/resources, quality/safety, material movements and accepted progress. Failed work generates accountable follow-up.
6. Assess / claim / pay	Costs & Contracts; package Money	Measured acceptance -> certificate -> invoice/bill -> payment. Separate commercial values, incurred cost and cash.
7. Test / hand over	Handover by floor/unit/system	Tests, snags, as-builts, O&M, training, warranties and acceptance. Partial handover has clear boundaries and outstanding obligations.
8. Defects / final settlement	Handover + Final account	Defects and warranty duties; agreed final account; authorised securities/retention release. Technical and commercial closure remain distinct.

The common context travels with the work

Company -> project -> work item/package + location/system. Link the programme activity, client BOQ line, subcontract scope, resource and evidence as relevant. A work breakdown and a physical location are different dimensions; neither should have to stand in for the other.

Changes, delays, approvals, records and cost control operate throughout these stages. A closed stage remains readable. New information can cause a controlled revision or reopening instead of silently overwriting the past.

Four journeys that expose weak mapping

Expert walkthrough of the proposed structure. These are reasoned checks, not observed user-test results or executed business transactions.

1. A failed concrete inspection

Site / inspection -> NCR -> authorised repair -> reinspection. Keep the inspection, location, cause, corrective actions and closure evidence together. Link labour/material/plant costs and waste to that work item. If a commercial change or delay is relevant, create a linked assessment; an NCR does not automatically become a billable variation.

Navigation test: the engineer can report the problem from the inspection; QA receives the decision in My Work; the PM sees the blocker. Nobody must find a separate approval or costing module to understand the case.

2. A subcontractor submits more than is accepted

Buying / package -> application -> assessed work -> certificate -> bill -> payment. Show claimed and accepted quantities with reasons for differences. The same package Money view explains deductions, retention, invoice status, due balance and payment. Finance acts on the linked authorised bill/payment record.

Navigation test: the QS and accountant can identify the held step and its owner without treating a certificate as a paid invoice. Partial/rejected work and revised claims remain traceable.

3. A drawing instruction affects time and cost

Documents / instruction -> change case -> linked delay assessment. The change owns scope, notice evidence, price and decision versions. Plan owns affected activities, mitigation and time analysis. Link both under the instruction; record pending/disputed status and accountable deadlines without double-entering the event.

Navigation test: both Plan and Costs & Contracts reveal the same originating event and the other assessment. A priced variation alone does not imply approved time or changed commitments.

4. The owner asks why one slab is overspending

Costs & Contracts / Budget & cost -> floor + slab item. Begin with revised budget, incurred cost, cash spent and forecast final cost at one cutoff. Expand material/labour/plant/subcontract/indirect detail, then source transactions. Show allocation gaps, productivity, waste, outstanding obligations and assumptions.

Navigation test: no separate search through stock, purchasing and finance is required to explain the totals. Source documents stay one link away, with their scope, tax basis and reconciliation status visible.

Every current menu has a home - 1

Traceability from the 42 observed administrator entries (E41). New destinations are proposals; repeated old labels do not prove duplicate functionality.

Current entry	Proposed destination	What is preserved or consolidated
01. Dashboard	My Work; Projects > Overview	Personal actions and project health have separate purposes; retire the generic duplicate entry.
02. Reports	Company > Reports; relevant workspace	One report definition; contextual views carry filters. Report setup belongs in Administration.
03. My Approvals & Activities	My Work	One decision/task queue, with explicit action, project, owner and due date.
04. CRM	Tenders > Opportunities	Keep leads, client follow-ups and bid decisions together. Contacts link to the shared directory.
05. Tender & BOQ	Tenders > Tender workspace	Tender documents, take-off, estimate revisions and bid review remain together.
06. Sales & Quotations	Tenders > Offers & award	Preserve quotation versions and accepted terms. Non-project sales, if needed, remain a named Company workflow.
07. Projects	Projects	Portfolio -> project overview -> stable project areas. Tasks appear in My Work and their project.
08. Project requisitions	Project > Buying > Requests	Site action Request materials opens the same request with location and work item filled in.
09. Project set-up	Project > Overview > Setup checklist	Mobilisation tasks, owners and missing baseline information; ongoing settings are secondary.
10. Material returns	Project > Site > Materials	Return from the original issue/receipt; shared warehouse view in Company > Stores.
11. WBS & Budgets	Project > Costs & Contracts > Budget & cost	Work items, cost codes, resource build-ups and approved versions; Plan links to the same work items.
12. Site receipts	Project > Buying > Orders & deliveries	Site > Materials offers Receive delivery; it opens the same delivery record.
13. Site diaries	Project > Site > Daily work	Daily report with labour, plant, progress, photos and events; do not re-enter shared facts.
14. Contracts & measured BOQ	Project > Costs & Contracts > Client contract	Award baseline, measurement rules and claims. Subcontract scope lives in its Buying package.

Project means the selected project workspace. A root-level map does not replace a later inventory of all 267 leaf controls, fields, reports, permissions and integrations before implementation.

Every current menu has a home - 2

Traceability from the 42 observed administrator entries (E41). New destinations are proposals; repeated old labels do not prove duplicate functionality.

Current entry	Proposed destination	What is preserved or consolidated
15. Site materials & issues	Project > Site > Materials	Available, reserved, issued, returned, installed and wasted quantities remain distinguishable.
16. Purchasing	Project > Buying; Company > Buying	Requests, comparisons, orders and deliveries. Company shows an all-project queue of the same records.
17. Payment certificates	Project > Costs & Contracts > Claims & payments	Client measurement/certification; supplier certificates stay within their subcontract package.
18. Advances & retention releases	Respective contract/package > Money	Client terms in Costs & Contracts; subcontract terms in Buying. Finance executes authorised settlements.
19. Final accounts & closeout	Project > Costs & Contracts > Final account	Commercial settlement linked to the separate Handover readiness record.
20. Guarantees & bonds	Respective contract/package > Securities	One register; expiry and release tasks in My Work. Finance has a company-wide securities view.
21. Subcontract finance	Project > Buying > Subcontract package > Money	Terms, certificates, bills, advances, retention and payment status. Posting remains in Finance.
22. Subcontract order amendments	Subcontract package > Changes	A linked amendment changes the commitment only when authorised; preserve its versions.
23. Subcontractor applications	Subcontract package > Work & claims	Application -> assessed work -> certificate; show differences and their reasons.
24. Subcontract service entries	Subcontract package > Work & claims	Accepted/rejected measured work; site reviewers get direct assigned tasks.
25. Subcontract work corrections	Original assessment > Correct or revise	A controlled correction with reason and history; no disconnected correction module.
26. Stores & Receipts	Company > Stores; Project > Site > Materials	Shared warehouse operations and project stock views. Deliveries link to Buying orders.
27. Labour & plant allocation	Project > Plan > Resources	Planned allocation; daily use captured in Site. Company People/Assets holds identities and availability.
28. Labour & plant actual feeds	Project > Site > Daily work; Administration > Integrations	Operational actuals are reviewable in context; technical imports/mappings stay with integration operators.

Project means the selected project. Contextual shortcuts open the same records; no duplicate entry.

Every current menu has a home - 3

Traceability from the 42 observed administrator entries (E41). New destinations are proposals; repeated old labels do not prove duplicate functionality.

Current entry	Proposed destination	What is preserved or consolidated
29. P6 schedule progress	Project > Plan > Programme	Progress, import freshness and baseline references. Preserve explicit programme ownership.
30. Finance	Company > Finance	Receivables, payables, cash/bank, ledger, close and reporting. Project views link the same facts.
31. Quality, HSE & Specifications	Project > Site > Quality & NCRs / Safety	Inspections, NCRs and safety cases; controlled specifications are linked from Documents.
32. People & HR	Company > People	Employee, recruitment and employment records; project assignments link here without exposing restricted details.
33. Time Off	Company > People > Leave; My Work	Employee request and manager decision are direct actions; balances and policy remain in People.
34. Attendance	Company > People > Attendance	Personal entry/correction and supervisor review; attendance is distinct from project costed time.
35. Timesheets	My Work > My time; Company > People > Timesheets	Project allocation is visible in Site; one approved time source prevents duplicate labour cost.
36. Plant & Equipment	Company > Assets & Fleet	Availability, custody, hire and maintenance. Project Plan/Site shows assigned plant and actual use.
37. Fleet	Company > Assets & Fleet > Vehicles	Vehicle-specific odometer, fuel, service and contract records remain explicit views.
38. Contacts & Masters	Company > Contacts & Catalogue	Shared parties, materials/services and units. Accounting and approval policy setup stays in Administration.
39. Team Discussions	Record > Discussion; My Work > Messages	Decisions attach to the relevant record; unlinked conversations remain searchable but do not approve work.
40. My Tasks	My Work	Merge the entry point with activities, keeping task type, source and completion rules.
41. Users & Access	Administration > Access	Explicit roles, company/project scope and delegated authority. No business module entry.
42. Administration	Administration utility	Company settings, policies, integrations, shared calendars and system operations; access restricted by duty.

Project means the selected project. Contextual shortcuts open the same records; no duplicate entry.

Twelve findability checks

Desk review performed against the proposed routes. No prototype exists and no click counts, timings or completion rates have been measured.

Task in the user's words	Proposed starting route	Information required to finish
Find today's overdue decisions	My Work > Needs my decision > record	Project, deadline, submitted version, evidence and allowed action; not an empty generic task dashboard.
Find an issued floor drawing	Project > Documents > Drawings	Filter floor/system; distinguish issue purpose, review state, current revision and superseded history.
Request and receive materials	Site action > linked Buying request/order	Project/item/location persists. Partial, rejected and returned quantities have explicit next actions.
Raise and close an NCR	Project > Site > Quality & NCRs	Source inspection, remedy authority and independent verification where required; follow rework cost.
Find a crane and its availability	Company > Assets & Fleet > asset	Location, custodian, service status and project allocation; maintenance block cannot be hidden by a calendar view.
Explain slab cost and spending	Project > Costs & Contracts > Budget & cost	Same item/cutoff: budget, incurred, paid and forecast. Expand resources and source facts without changing context.
Check a subcontractor's unpaid claim	Project > Buying > package > Money	Claimed/assessed/certified/billed/due/paid values and owner of the held step; linked Finance record.
Assess a design change and delay	Drawing or Site event > Raise change	One case references instruction, affected work, notice, commercial decision and linked delay/EOT assessment.
Explain next month's cash gap	Project > Costs & Contracts > Cash flow	Expected dates, overdue receipts, approved obligations and scenario assumptions; company view retains source scope.
Record waste and recover its cost	Project > Site > Waste	Measured material loss versus disposal movement, reuse, evidence and allocation; no double-counted material cost.
Prepare partial floor handover	Project > Handover > location/system pack	Outstanding tests/snags, current as-builts/O&M and acceptance evidence; contractual gate exceptions are authorised.
Reconstruct who approved a change	Change > History; search by ID	Version, actor, decision, reason and linked effects; audit history is distinct from a financial fact register.

Review result: each of these tasks now has a named home and a linked handoff. Remaining hypotheses include whether Company is clear enough, whether users expect changes under Costs & Contracts, and whether snag discovery works equally well from Site and Handover. Test these first.

Prove the map before building it

Review conclusion: this is a stronger, traceable proposal. Calling it the best user experience requires evidence from representative users.

What the second review changed

Removed competing top-level module choices; split vague Resources into named Company workspaces; gave Documents and Handover permanent homes; defined package ownership and Finance handoffs; removed stage-dependent navigation; mapped all 42 roots and all twelve requested capabilities. Richness now comes from connected records, not more primary menu entries.

A small, measurable validation programme

Round	Method and proposed acceptance target
1. Labels and tree test	Give realistic tasks without menu words. Test the proposed map with you and representatives of site, QS/PM, buying, quality/documents and finance; add estimator/plant/people users for their routes. Target at least 90% correct first destinations across core tasks; inspect failures by role, not only the overall average.
2. Paper/clickable task walkthrough	After study review, test current drawing, NCR, partial delivery, subcontract claim, item cost, change/delay and handover cases. Target at least 90% unassisted completion and median information-finding time under 60 seconds. These are proposed gates, not achieved results.
3. Comprehension and exceptions	Every participant handling money must correctly distinguish budget, incurred, cash spent and forecast on the tested item. Critical tested decisions must have the right version, authority and scope. Include wrong project, missing allocation, rejected work, stale data, partial payment and reopening.
4. Accessibility and repeat use	Test keyboard/focus, zoom, touch and relevant assistive technology; mobile capture and poor connectivity where needed. Repeat with revised labels and different examples; check return visits without coaching.

Record the baseline and agree targets before testing; revise the design when it fails. Do not average away a severe control failure or conceal poor results for one role. Test browse routes as well as search, and record wrong turns and repeated entry.

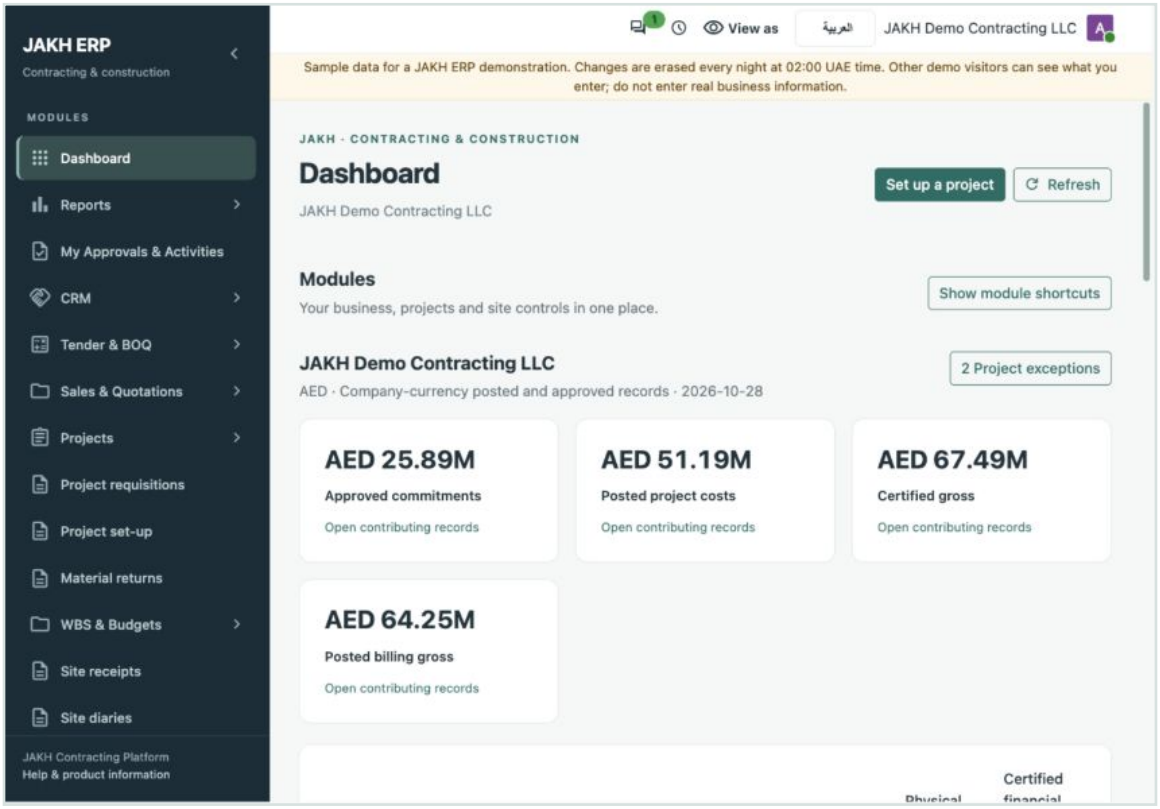
Principles checked against published guidance

Nielsen Norman Group: Progressive Disclosure supports keeping frequent needs visible and clearly labelling secondary detail. W3C: Consistent Navigation supports predictable relative ordering. GOV.UK: Task list distinguishes flexible task completion from a strictly ordered process. These support the principles, not the exact four/seven-area grouping. Accessed 28 October 2026.

Navigation needs a new structure

Evidence: E39 administrator dashboard, E41 expanded navigation inventory, E30/E50/E53 role dashboards.

Role	Primary entries	What this proves
Administrator	42	Broad modules and individual transactions share one level.
Project Manager	20	Role filtering still leaves a long operational menu.
Site Engineer	10	A smaller role-specific menu already exists.
Estimator	9	Tender work can be presented with fewer primary areas.



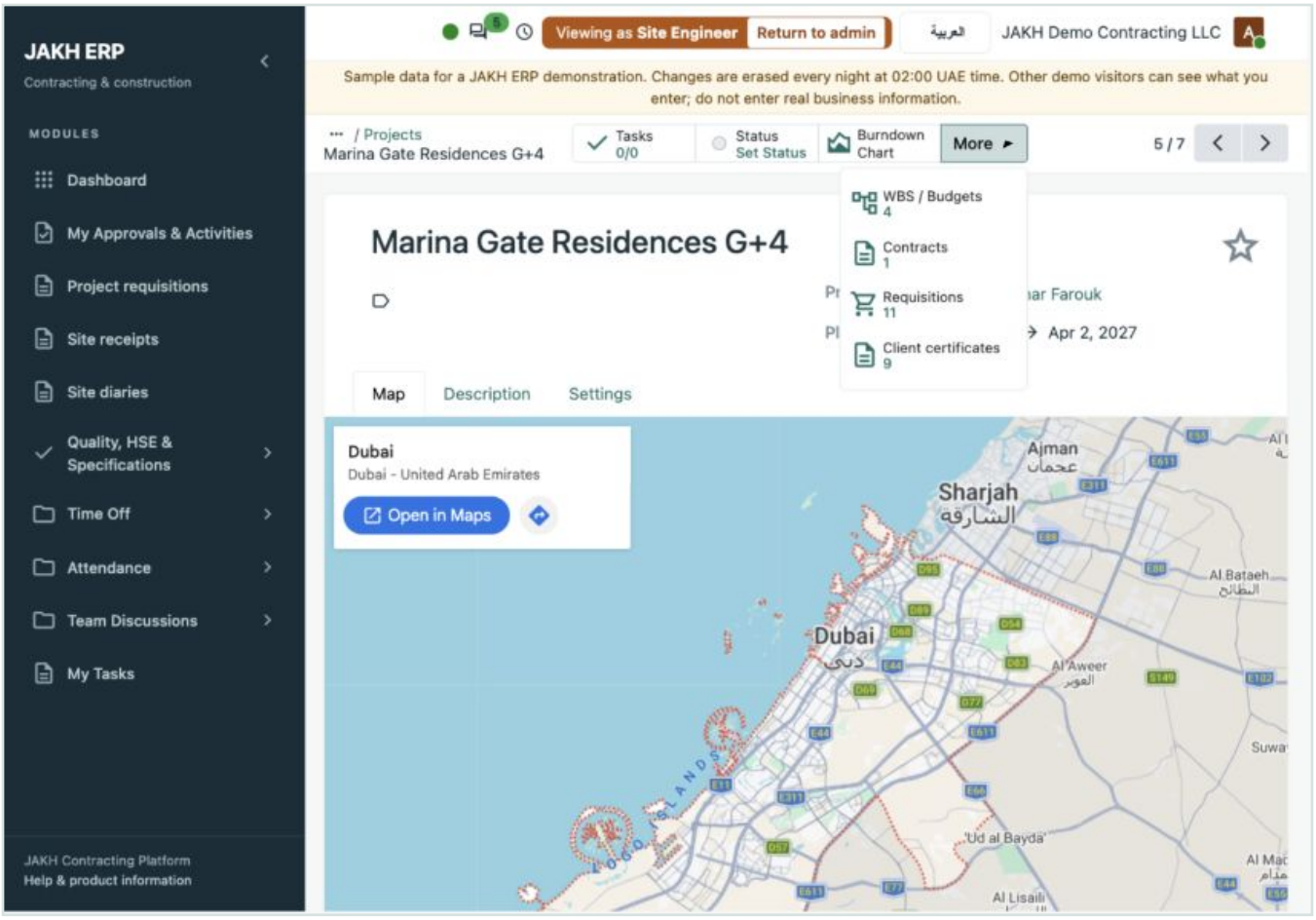
E39. Administrator in JAKH Demo Contracting LLC. The screenshot shows the first portion of the scrollable sidebar.

The fully expanded administrator tree contains **363 button controls: 96 groups and 267 leaf buttons**, with paths reaching four menu levels. Leaf counts are navigation controls, not a count of unique features.

Repeated leaf labels include Products (6), Customers (4), Settings (12), Dashboard (4), Payments (3), and My Tasks (2). Their targets were not all compared; identical labels do not prove duplicate records or duplicate functionality.

A project should open on work

Evidence: E32 project entry and E33 its More menu, viewed as Site Engineer.



E33. Marina Gate Residences G+4 opens on Map / Description / Settings. WBS, Contracts, Requisitions and Client certificates sit under More.

Observed friction

The main project view prioritises a large Dubai map, generic task counts, and project metadata. It does not immediately answer what is happening on site, what is late, what needs approval, or which floor/package needs attention.

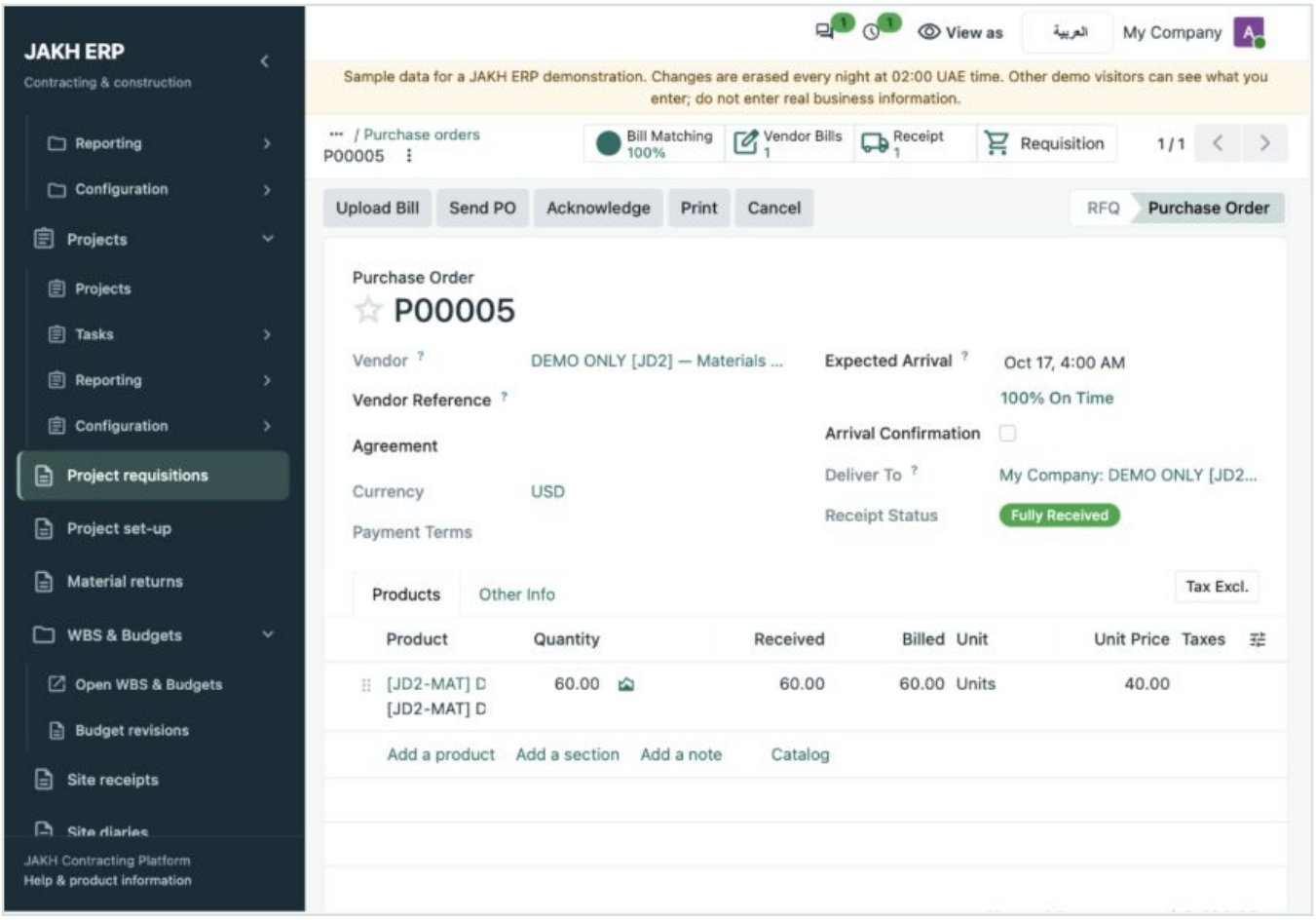
Requirement to validate

Use a project workspace with a stable identity and location/package filter. Put current actions, blockers, recent progress, and relevant documents within that context. Treat maps and administrative settings as supporting information.

Waiting records should name the next owner and expected review date, including when the viewing user cannot approve. The absence of an approval button for a site role may be correct permission behaviour.

Keep the procurement chain connected

Observed path: requisition -> linked PO list -> purchase order -> completed receipt. Evidence E19-E22.



E21. P00005 shows received/billed quantities and links to the receipt, bill, and requisition. Its main view does not give project/WBS/location the prominence seen on the requisition.

What works, and where context weakens

The requisition includes the project, WBS, required date, planning-demand reference, and requested location. The PO exposes the vendor, arrival, receipt status and quantities. The receipt exposes the source PO and stock movement. The linkage is present; the user must reconstruct its construction context.

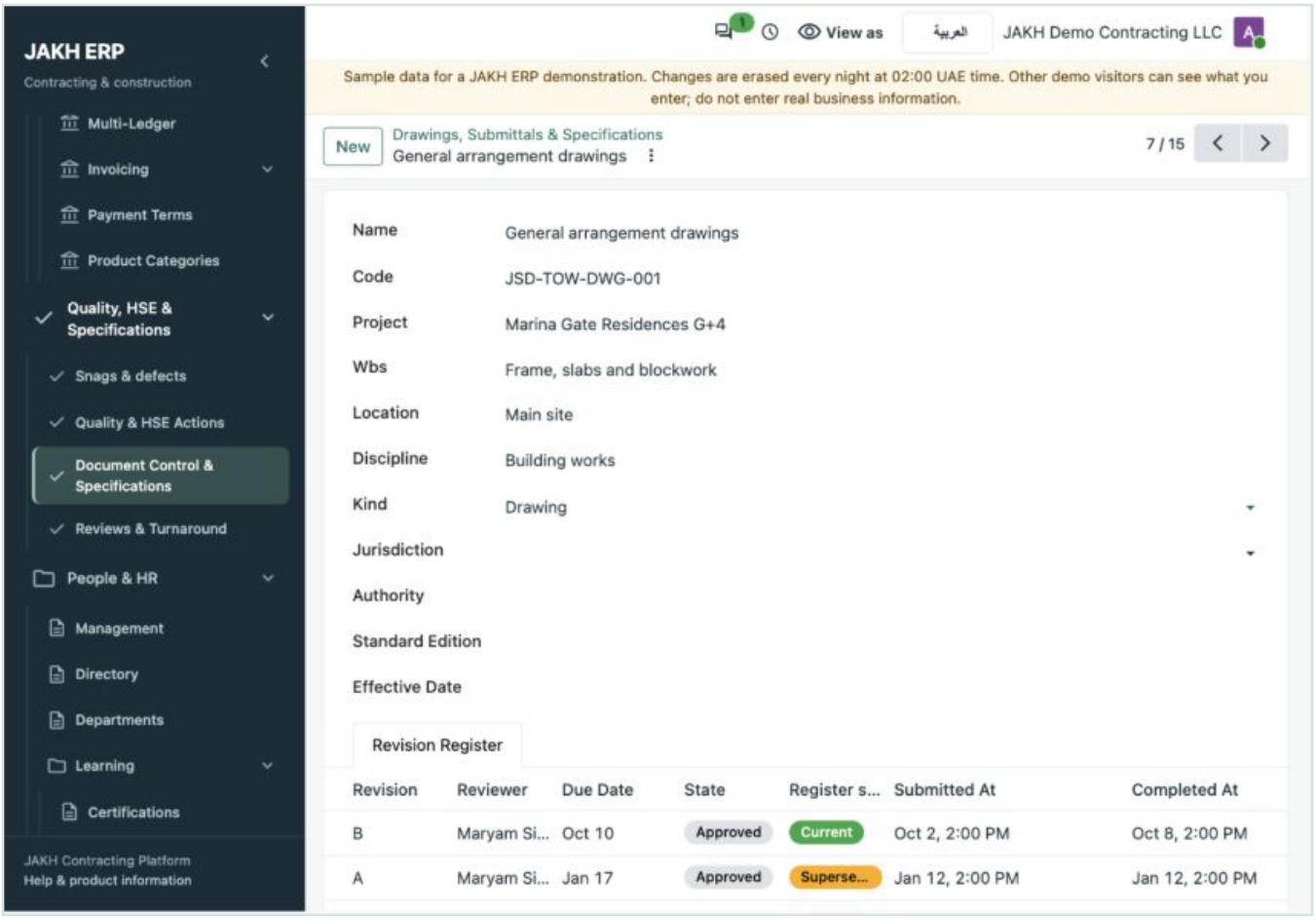
Requirement to validate

Carry project, package and location visibly across demand, request, quotation, order, receipt, inspection, issue, installation, and invoice. Link distinct controlled records rather than merging them into one editable document.

In this sample, a link labelled 1 Purchase orders opens an intermediate single-row list before the PO. Test direct opening when there is exactly one related record. Partial delivery, rejection, returns and billing exceptions were not executed.

Technical controls exist

Evidence: E43 document register, E44 revision history, E45 document-category dropdown.



E44. General arrangement drawing JSD-TOW-DWG-001 shows approved revision B as Current and approved revision A as Superseded.

Preserve revision meaning

The current-versus-superseded distinction is useful. A register also needs to let a site user find the correct approved document for the floor, zone, trade and activity without searching through unrelated projects.

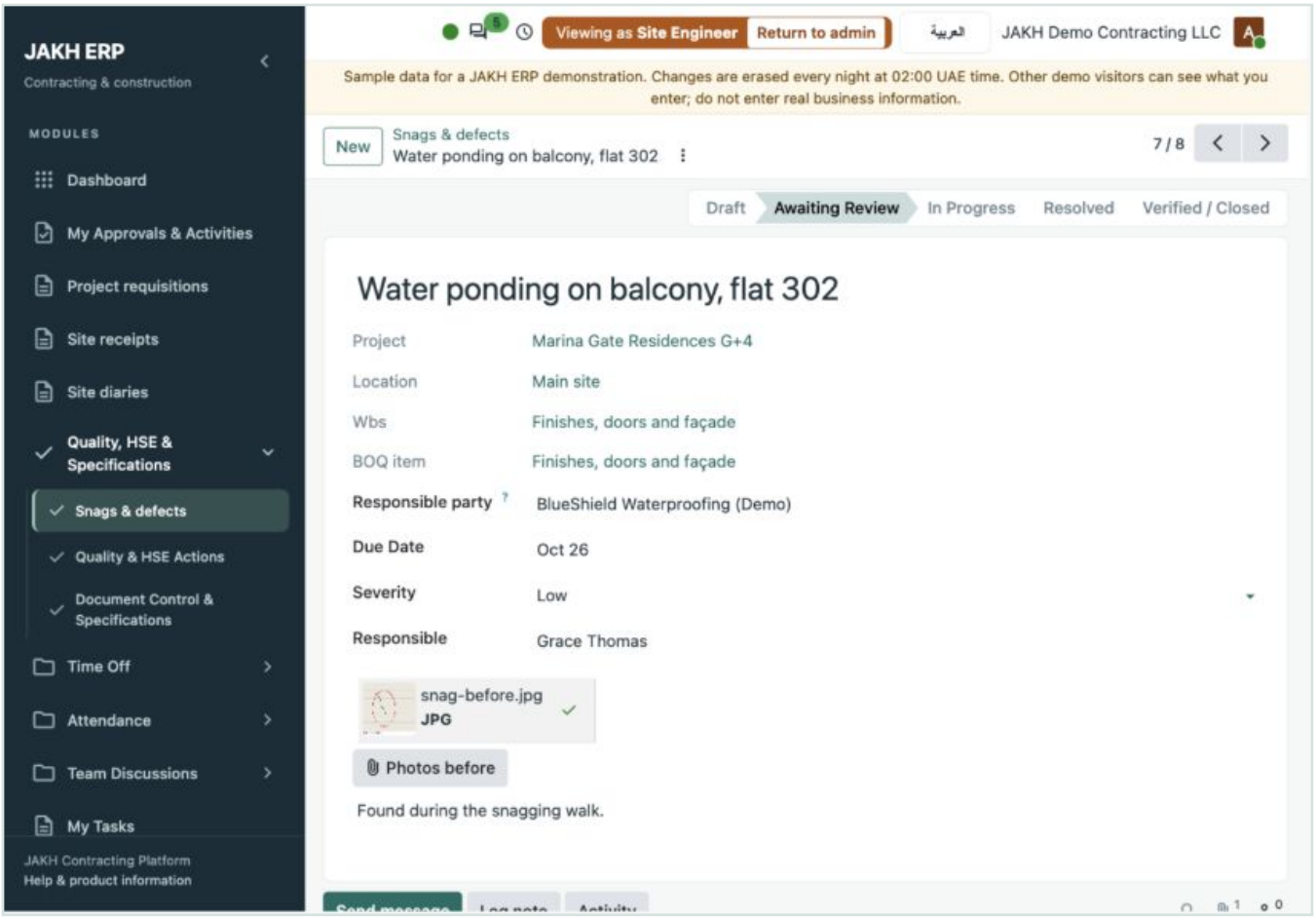
The category control offers Drawing, Submittal, Specification, Method Statement, Inspection / Test Plan, and Correspondence. The fuller register contained 15 drawing records; other category workflows were not demonstrated by populated samples.

Still to validate

Dedicated RFI routing, drawing release/transmittal, consultant response codes, inspection hold points, document distribution and external-party review. A category called Correspondence does not by itself demonstrate an RFI workflow.

Tower work needs structured locations

Evidence: E34 snag register, E35 snag detail, E37-E38 submitted site diary.



E35. Water ponding on balcony, flat 302 is linked to a project, WBS, BOQ item, contractor and person. Its structured Location is Main site.

A concrete retrieval problem

The eight displayed open snags all use Main site, including titles referring to a flat or individual villa. For these records, finding work by floor or unit depends on free text. This is a sample-data limitation and an information-structure risk; it does not prove the application cannot represent locations.

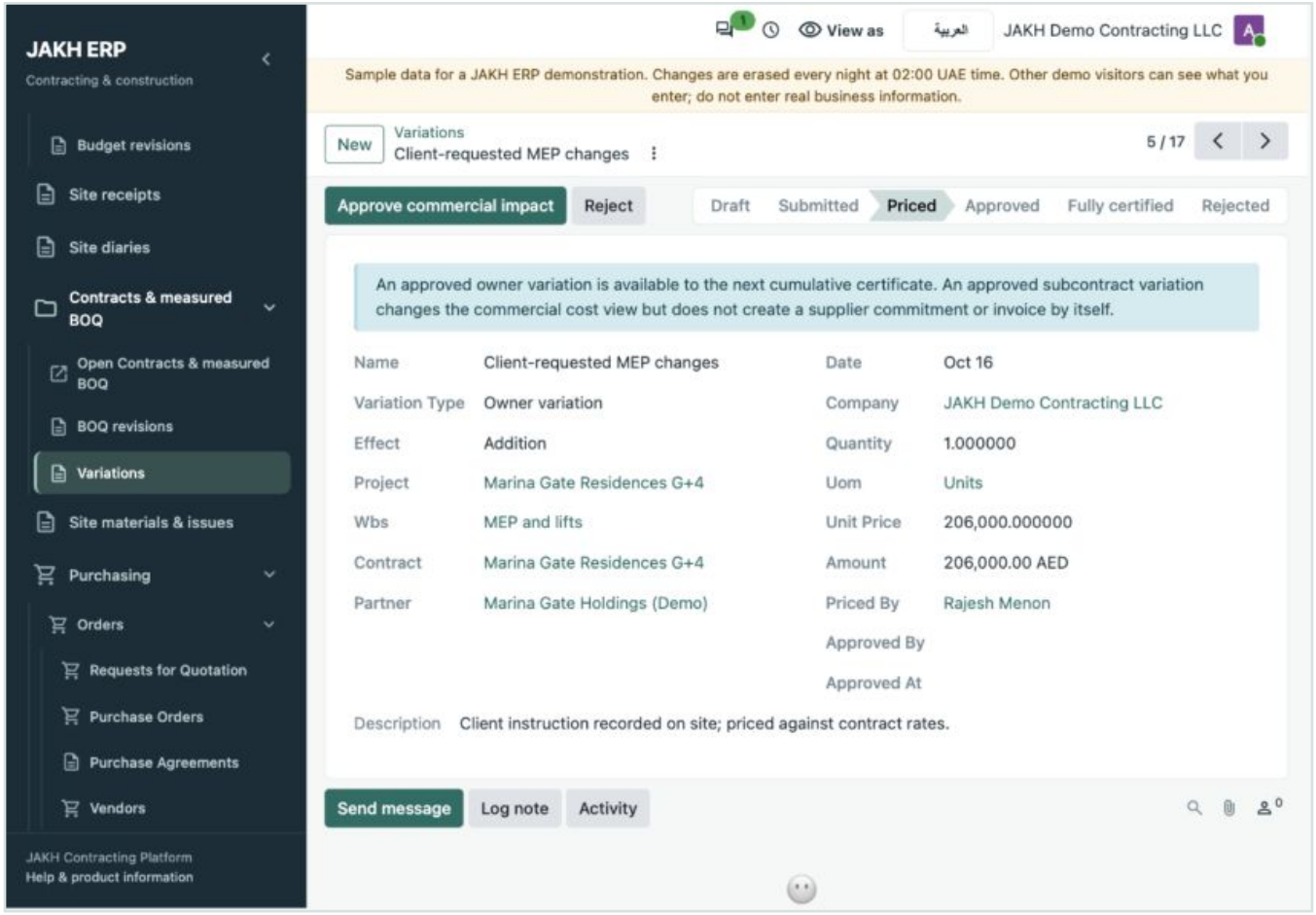
Requirement to validate

Model project -> tower/building -> floor -> unit/zone, with systems and packages linked separately. Test bulk snagging, photos, contractor assignment, rectification, reinspection, reopening and partial handover by location.

The site diary has purposeful Manpower, Equipment, Photos, and Work and events tabs. Keep useful grouping. Its Submitted state and the snag's Awaiting Review state need clearer next-review ownership in the captured views.

Connect changes to their consequences

Evidence: E46-E47 variation records; E15 client certificate and E29 subcontract service entry.



E47. A priced MEP variation has project/contract links, quantity, rate, value, pricing owner, and approval actions. No instruction link, notice deadline, or time impact is visible on this record.

Commercial coverage is present

The sampled client certificate contains cumulative quantities, prior/current/remaining measurements, approved variations, retention/recovery values, an invoice link, and technical/commercial/finance review history. A submitted subcontract service entry also has package, location and quantity context.

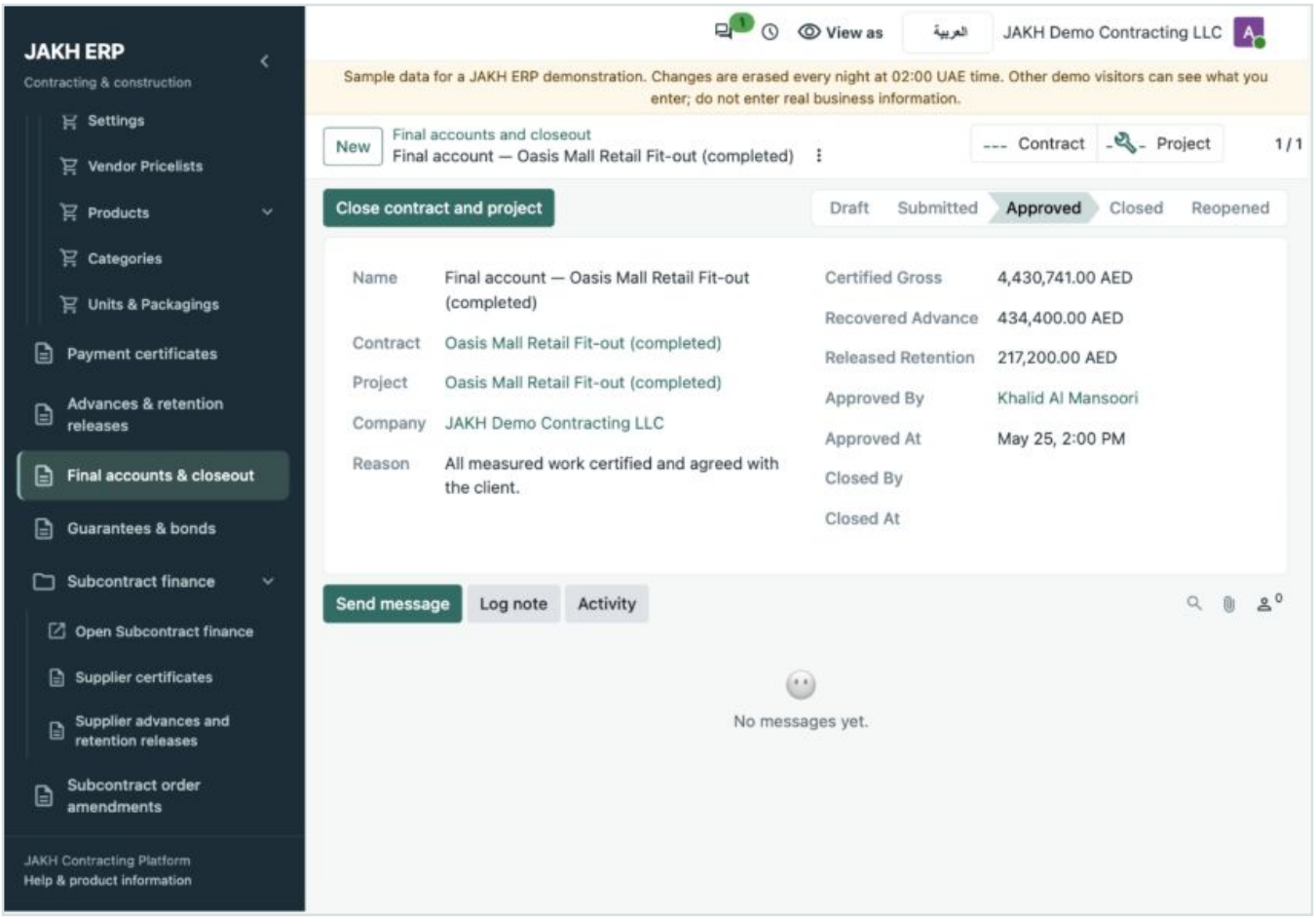
The chain to validate

Source instruction -> required notice -> estimate and cost/time impact -> review -> approval or dispute -> budget/programme change -> measured work -> certification -> invoice -> cash. Include work proceeding before approval and rejected or partially certified amounts.

Missing fields on the sampled variation are not proof of system-wide absence. Contract-specific notice rules, approval authorities and finance treatment require validation with your business owners.

Handover is more than a final account

Evidence: E48-E49 final-account workflow; E34-E35 snagging. No closure action was executed.



E49. The completed retail-fit-out project has an Approved final account and a Close contract and project action. The visible detail focuses on certification, advance recovery and retention.

Separate three completion questions

Technical readiness: are inspections, testing/commissioning, authority requirements, as-builts, O&M manuals, warranties, training and required snags complete?

Client acceptance: which building, floor, unit or system is accepted, by whom, with what outstanding obligations and defect-liability dates?

Commercial settlement: are measurements, variations, claims, subcontract settlements, retention and final balances agreed?

The sampled final-account screen does not visibly present a structured technical-readiness pack. Backend closure checks and wider handover workflows were not tested. Define the three tracks explicitly before designing the replacement.

Construction coverage map - 1

Observed means visible in this run. Not demonstrated means unverified, not proven absent. Execution/approval transitions were not performed.

Lifecycle area	Current evidence	Requirement / next verification
Tender pipeline / bid decision	Observed: generic CRM entry; estimator preview has no matching opportunities (E51).	Validate bid/no-bid, tender deadlines, qualifications, clarifications, tender security and opportunity access.
BOQ / estimate / revisions	Observed: estimates, study versions, pricing, review and transfer links (E05, E52).	Validate take-off detail, quote comparison, exclusions, alternate bids, revision comparison and approval limits.
Award / project mobilisation	Observed: estimate-to-quotation/project transfer; setup entry (E05-E07, E26).	Validate contract obligations, responsibilities, programme and budget baseline; direct-award path.
Project / tower locations	Observed: location records; fuller building sample uses Main site (E35, E44).	Validate tower/floor/unit/zone hierarchy and large-volume navigation. G+4 is not a high-rise scale test.
Budget / cost codes / forecasts	Observed: BOQ/location cost report, source links and approved cost-to-complete forecasts (E66-E69, E90).	Validate item/resource detail, revisions, commitments, incurred costs and cash reconciliation. See pages 29-31.
Programme / look-ahead	Observed: 145 progress records; the detail says P6 is schedule master (E79-E80).	Validate activity logic, baselines, imports, look-ahead and delay/EOT linkage. See page 34.
Drawings / submittals / documents	Observed: current/superseded drawings and six categories (E43-E45).	Validate submissions, transmittals, distribution, consultant response and overdue review ownership.
RFIs / instructions	Not demonstrated as a dedicated workflow in the inspected views.	Trace RFI -> response -> drawing/work instruction -> cost/time consequences. Do not treat a label as proof.
Materials / procurement	Observed: demand/request/order/receipt links (E19-E22).	Validate quote comparison, approvals, lead times, partial/rejected delivery, returns and material inspection.
Stores / site issues / installation	Navigation and demand links observed; full operational chain not walked.	Trace stock source, reservation, transfer, issue, installed quantity, wastage and cost allocation.
Subcontract award / amendments	Navigation and subcontract order/service links observed (E28-E29, E41).	Validate package scope, exclusions, measurement rules, changes, commitments and subcontract securities.
Subcontract measurement / payment	Observed: service entries, supplier certificate, bill and retention agreement links (E29, E70-E72).	Validate application -> acceptance -> settlement; rejected work, deductions and correction. See page 28.

Construction coverage map - 2

This map is a requirements register for the new ERP. Supporting functions remain in scope even when sampled only through navigation.

Lifecycle area	Current evidence	Requirement / next verification
Daily reports / labour / plant	Observed: populated diaries with resources and work/events (E36-E38); allocation/feed entries.	Validate who submits/approves, location/package detail, hours, equipment use and cost-feed reconciliation.
Quality / HSE / inspections	Observed: non-conformances, inspections and risk assessments; NCR resolution/lifecycle (E56-E59).	Validate ITP hold points, cause/corrective action, independent verification and safety controls. See page 25.
Variations / notices / delay	Observed: price/approval states; time/notice chain not visible in sample (E46-E47).	Define instruction evidence, notice dates, cost/time assessment, disputed work and linked approval history.
Client measurement / certification	Observed: cumulative BOQ/variation measurement and staged approval (E15).	Validate claimed vs certified/rejected values, evidence, corrections, prior quantities and invoice progression.
Finance / cash / accounting	Observed: bill and matched receipt sources. Cash report blocked by policy selection; visible policy list empty (E70, E74-E76, E91-E92).	Validate forecasts, policy setup, reconciliation and original posting/localisation. See pages 27 and 30.
Approvals / delegation / exceptions	Observed: role routes/history; PM activity queue has two threshold tasks (E15-E16, E54).	Verify queue coverage for all pending documents; delegation, absence cover, rejection/resubmission and independence.
Testing / commissioning	Not demonstrated as an end-to-end workflow in inspected records.	Define system/asset tests, dependencies, witnesses, results, failures/retests and readiness evidence.
Snagging / rectification	Observed: populated snags, owners, before photo and status sequence (E34-E35).	Validate after-photo requirements, independent verification, reopening, bulk capture and unit/system filtering.
Handover / defect liability	Not demonstrated as a structured handover pack in the sampled final-account view (E49).	Define partial acceptance, as-builts, O&M, warranties, training, DLP dates, outstanding work and responsibility.
Final accounts / retention	Observed: financial final-account workflow; sample Approved and Closed records (E13, E49).	Validate technical/contractual readiness gates, settlement, retention release and auditable reopening.
People / assets / shared masters	Observed: equipment, serial/cost and maintenance detail (E61-E64). Most HR/fleet scope remains navigation-only.	Validate asset custody, utilisation, hire, finance and handover; retain required HR/shared scope. See page 26.
Portals / integrations / migration	Role chooser exposes client/subcontractor portals; schedule and cost feeds have entries.	Verify portal permissions, external reviews, integration ownership, data/attachment export, migration and cutover.

Your 12 areas: existence and need

Expanded scope map 1 of 2. Follow-up inspection: administrator, JAKH Demo Contracting LLC, AED, 28 October 2026.

All twelve belong in the construction ERP scope. The depth varies, but none should disappear merely to shorten the menu. Place them inside coherent project workspaces and shared services. A visible record proves that capability exists at that level; it does not prove a complete or reliable end-to-end process.

Area / detail page	Current evidence and status	Need in the new ERP
1. NCR operations Page 25	PARTIAL: populated Non-conformance records, resolution, verifier and timestamps. E56-E59.	Essential from mobilisation: contain defective work, resolve the cause, verify the remedy, and trace rework consequences.
2. Asset management Page 26	PARTIAL: 20 equipment records; crane serial/cost, maintenance links and reliability metrics. E61-E64.	Essential: know what is available, where it is, who has it, whether it is fit for use, and what it costs. Separate owned/hired plant and client handover assets.
3. Payments & cash flow Page 27	PARTIAL / BLOCKED REPORT: posted receipts and bills exist. Company cash report requires a policy; visible policy and cash-forecast lists are empty. E70, E73-E76, E91-E92.	Essential: expected versus actual money, approved obligations, collections, payment scheduling, advances, retention and liquidity.
4. Subcontracting Page 28	SUBSTANTIAL PARTIAL CHAIN: service-entry/certificate/bill links, finance agreement, retention terms and balances. E29, E70-E72.	Essential: manage each package from selection and award through measured work, changes, payment and final settlement.
5. Item cost breakdown Pages 29-31	PARTIAL: BOQ/location report, source facts, WBS budgets and approved cost-to-complete forecasts. Samples remain coarse. E66-E69, E87-E90.	Essential: one explainable item ledger with budget, quantities, resources, incurred cost, commitments, payments and completion forecast.
6. Waste management Page 32	LIMITED: empty scrap register and unsaved stock-scrap form. Full site waste chain not demonstrated. E77-E78.	Essential: material loss and environmental disposal/reuse records; costs and quantities linked to the work that caused them.

Status key: PARTIAL = useful records or controls observed, with workflow depth unverified. LIMITED = narrow or unpopulated evidence. BLOCKED REPORT = a specific report could not produce results under the inspected setup. None means proven absent throughout the system.

Your 12 areas: controls and visibility

Expanded scope map 2 of 2. Proposed requirements below are design inputs, not claims about existing functionality.

Area / detail page	Current evidence and status	Need in the new ERP
7. Variations / change requests Page 33	PARTIAL: 17 variation records; owner/subcontract distinction, pricing, approval and certification states. E85-E86.	Essential: capture an early change event, assess scope/cost/time, obtain authority, and carry the outcome into budgets, orders and programme.
8. Programme & delay Page 34	PARTIAL: accepted progress facts; screen explicitly makes P6 the schedule master. Delay/EOT chain not demonstrated. E79-E80.	Essential: baseline/current plan, look-ahead, constraints, delay events, mitigation and time-impact evidence.
9. KPI management Page 35	PARTIAL: CPI/SPI, project financial/progress metrics, exception activities, turnaround data and generic dashboards. E53-E54, E66, E82-E84.	Essential: definitions, owners, targets, trends, data freshness and corrective actions; a chart alone is insufficient.
10. Keeping records Page 36	PARTIAL: revisions, attachments, diaries, lifecycle rows and source links. E37-E45, E57-E59, E82.	Foundation: authoritative evidence, controlled revisions, search, distribution, retention and export throughout the project.
11. Approval workflow Page 37	PARTIAL: staged certificate review, named forecast approver, quality verifier, document review and activity queues. E15, E54, E59, E71, E82, E90.	Foundation: consistent, policy-based routing with next owner/due date, rejection/resubmission, delegation and version-aware decisions.
12. Tracking / transparency Page 38	PARTIAL: project facts link to source records; dates, provenance and some approver fields exist. E69-E72, E81-E82, E90-E92.	Foundation: show where work stands, why it is blocked, who acts next, and how every total was derived, within access rights.

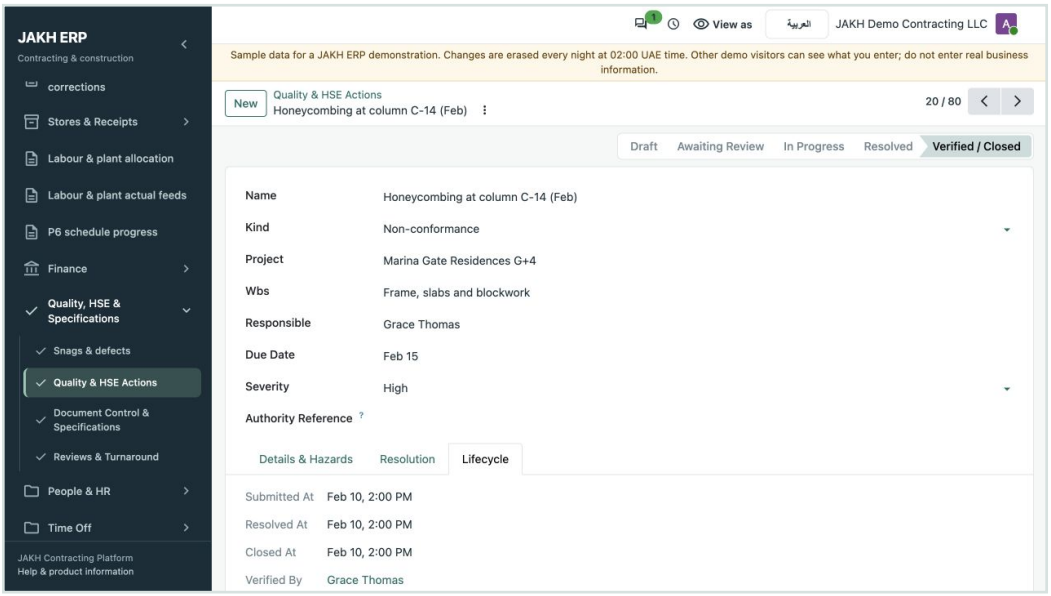
How these fit together

The common spine is company -> project -> building/floor/zone -> work item/package, with linked programme activity, resource, party and evidence. Records, approvals and tracking serve every workflow. Quality, cost and schedule consequences must travel with the originating event.

Example: failed inspection -> NCR -> authorised repair -> labour/material/plant use and waste -> reinspection -> accepted progress. If scope or time changes, link the change/delay assessment. Do not assume every NCR automatically creates a billable variation.

1. NCR operations

Existence: confirmed at record level; completeness: partial. Route: Quality, HSE & Specifications -> Quality & HSE Actions. E56-E59.



E59. A closed honeycombing NCR has project/WBS, responsibility, severity, lifecycle dates and a named verifier. Resolution is a separate free-text section.

What exists; what remains uncertain

The register mixes inspections, non-conformances and risk assessments. The sampled NCR has one attachment and a generic correction statement. Its responsible person and verifier are the same in this sample. Structured requirement breach, root-cause analysis, distinct corrective-action tasks, effectiveness review and linked rework costs were not visible; independence rules were not exercised.

Needed workflow and simple UI

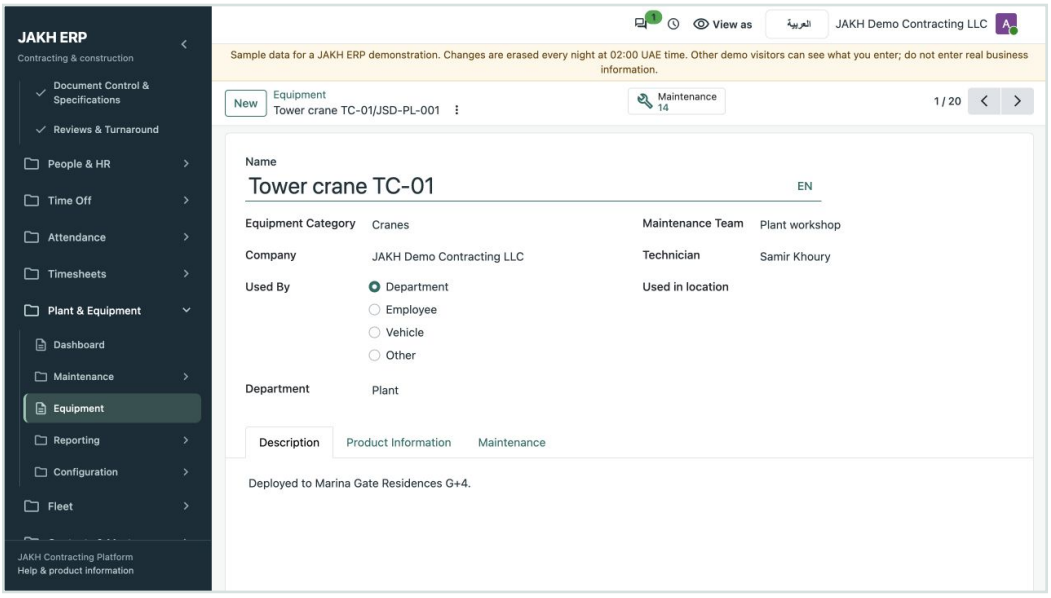
Essential. QA/QC raises the issue against an inspection, specification and exact location; the site team contains it; the responsible contractor proposes a disposition; required technical/client reviewers authorise it; the team implements and supplies evidence; QA/QC verifies and closes or reopens. Record cause, prevention, owners, deadlines and cost/time impacts. Keep it in Project -> Site -> Quality & NCRs, with one case timeline and the action required now.

Test: trace one rejected concrete item through repair, reinspection and linked costs; show why it may be closed. Where policy requires independent verification, prove the resolver cannot provide that approval.

Reference reviewed: the public QIC NCR toolkit description covers a form, procedure, register and dashboard, including cause analysis, corrective action and effectiveness checks. Paid files were not acquired or inspected. This is a process reference, not certification evidence or a copied template.

2. Asset management

Existence: equipment and maintenance confirmed; a complete asset lifecycle is not demonstrated. E61-E64.



E62. Tower crane TC-01 has a maintenance team and 14 linked maintenance records. The project deployment is in description text; Used in location is blank.

Preserve, then connect

Twenty equipment records are visible. The crane has a serial number, cost, vendor/model/warranty fields and reliability measures. Fleet also has navigation entries. These do not establish fixed-asset accounting, hire control, transfer history, inspection/calibration validity, fuel/use reconciliation or client asset handover.

Needed workflow and simple UI

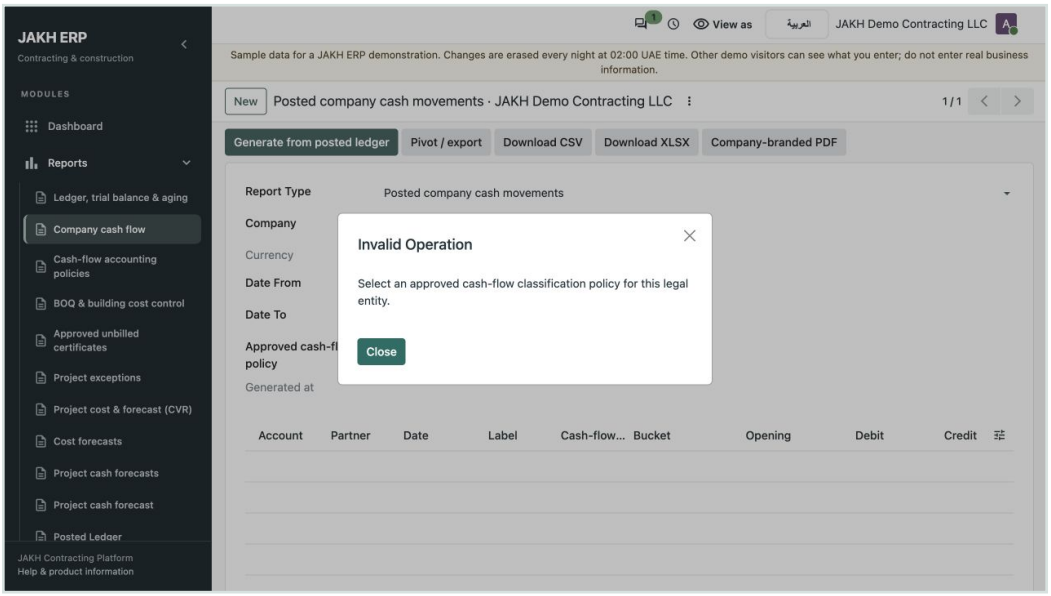
Essential, with three related tracks. Owned plant/tools: acquire -> identify/tag -> assign/transfer -> use -> maintain -> dispose. Hired plant: hire terms -> on-hire -> utilisation/downtime -> off-hire -> invoice check. Installed client assets: install -> test/commission -> warranty/O&M -> handover. Link rather than duplicate asset identities across operations and finance.

The plant manager owns availability and service dates; site custodians own location/use; finance owns valuation and depreciation policy. Link hours, fuel, repairs, hire, depreciation or approved internal charges to the appropriate project/work item. Keep Company -> Assets & Fleet as the shared register, and show assigned assets in each project.

Test: transfer a crane between projects, stop scheduling it while out of service, reconcile its utilisation/cost, and find its inspection evidence. Separately retrieve one installed pump's commissioning and warranty pack.

3. Payments & cash flow

Existence: payment evidence confirmed; forecast depth unverified; company cash report blocked by its policy prerequisite. E70, E73-E76, E91-E92.



E74. Company cash generation requires an approved classification policy. E75 shows an empty visible policy register; E76 shows no project cash-forecast records.

What the evidence supports

The building report links to eight posted payment sources totalling AED 15,675,795.22. One source has balanced bank/receivable journal lines and an invoice reference. The report explicitly does not assert bank clearance. A subcontract bill shows AED 257,484 gross work, AED 25,748.40 retention and AED 231,735.60 due. Certification, invoicing, due balances and cash are different facts.

Needed workflow and simple UI

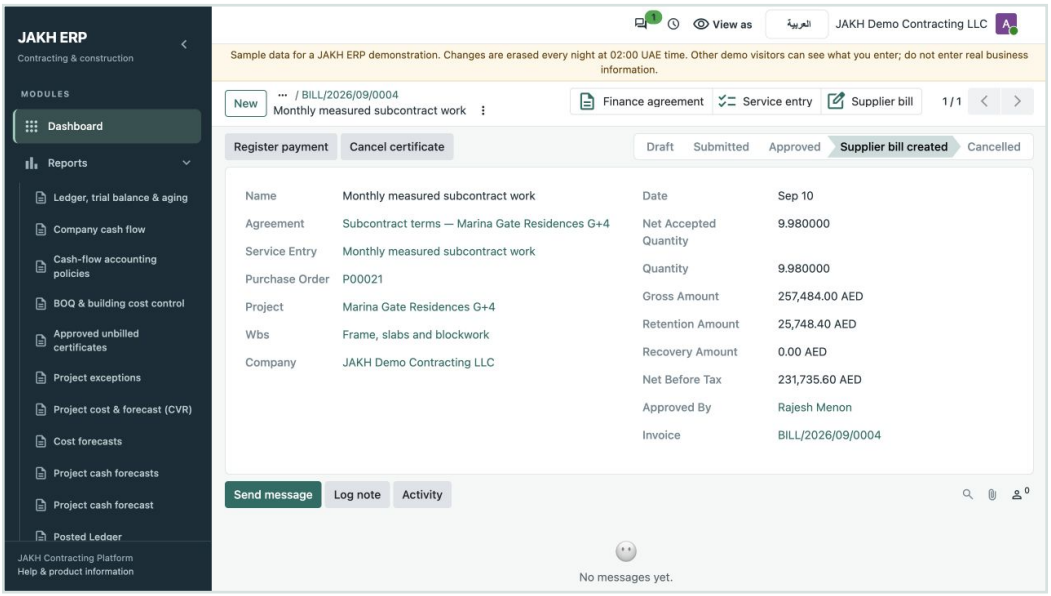
Essential. Finance and QS maintain expected receipts and outflows from payment terms, certificates, invoices, payroll, orders, taxes, advances and retention. Use a rolling weekly view for near-term liquidity and monthly project-to-completion outlook, with base/late-payment scenarios. Reconcile actual bank movements, forecast dates and invoice allocations; explain unmatched or overdue items.

Show project cash in Project -> Costs & Contracts, and the same records in Finance for company-wide scheduling and reconciliation. Display planned, approved, due, paid/received and bank-reconciled states separately. Include approval and bank authority checks; report access must not imply payment authority.

Test: follow partial settlement, advance recovery, retention release and a late receipt through both project and company views. Validate the cash-policy setup before relying on the company report. No payments were initiated in this study.

4. Subcontracting management

Existence: a substantial measurement-to-bill chain is visible. Award, exceptions and settlement remain to be validated. E29, E70-E72.



E71. The supplier certificate connects service entry, PO, project/WBS, finance agreement, approver, retention and bill.

Current strengths and gaps

The linked agreement holds retention percentage/cap, advance/recovery terms, balances and eight certificates. The bill links back to the service entry, PO and supplier certificate. Separate application, amendment and correction menus exist. Tender comparison, scope boundaries, securities, insurance expiry, disputed measurements, backcharges and final settlement were not walked end to end.

Needed workflow and simple UI

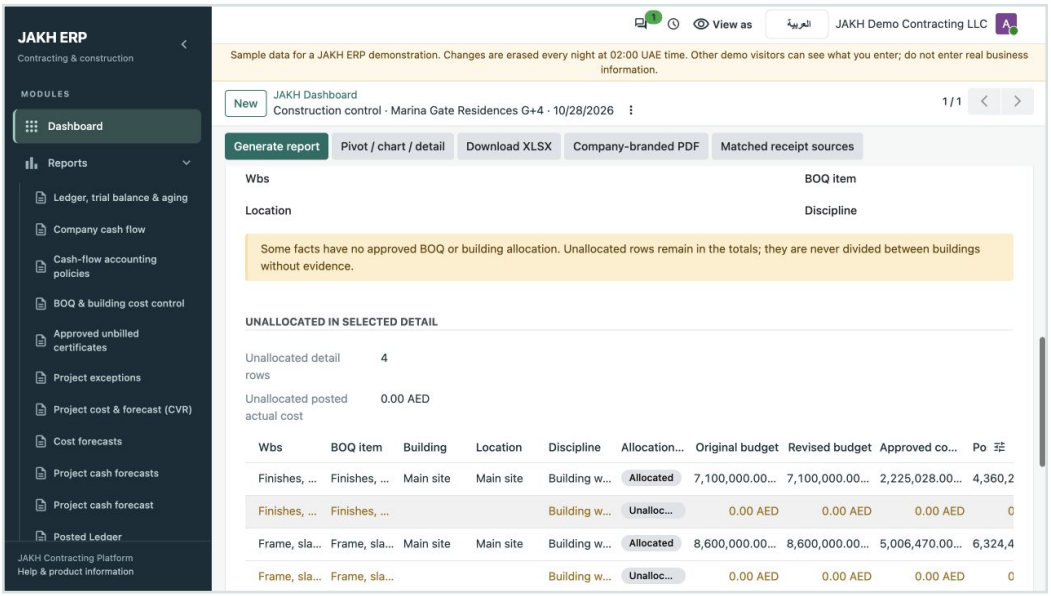
Essential. Procurement/QS prequalify and compare bids -> approve package scope and award -> record mobilisation obligations -> receive the subcontractor's application -> measure/inspect accepted work -> assess deductions and changes -> approve certificate -> match bill -> pay -> settle and release retention against agreed conditions.

Use one package workspace: scope/BOQ, commitments, progress, changes, quality obligations and money. Keep claimed, assessed, certified, invoiced and paid amounts distinct. Link NCRs, rectification, securities and handover readiness. Company -> Buying provides cross-project queues; project and portal views reference the same authorised records.

Test: certify only accepted quantities from a partly rejected application, amend scope, apply an evidenced deduction, then reconcile the package account without counting both the service receipt and bill as separate incurred cost.

5. Cost analysis for each site item

Existence: BOQ/location cost reporting and source lineage are confirmed. Required tower/resource granularity is not yet demonstrated. E66-E69, E87-E90.



E67. The report distinguishes original/revised budgets, commitments and posted cost, with BOQ/building/location fields and explicit unallocated rows.

A useful foundation with a granularity gap

The sampled building uses four broad WBS/BOQ groups, generally allocated to Main site. This does not demonstrate the cost of a particular floor slab, door type or installation activity. Source facts distinguish commitments, received exposure, posted cost and revenue/billing. Approved forecasts include actual-cost snapshots and remaining costs; report rules avoid silently dividing WBS forecasts among buildings.

One WBS form shows Forecast Amount = 0 while its dated approved forecast is AED 8,428,000 (E88-E90). The report says it uses the approved forecast. The UI should identify that authoritative value and explain the other field; the difference is not proof the report calculation is wrong.

Required grain and ownership

Essential. QS/cost control defines work items and cost codes; planning supplies quantities and dates; site/stores supply consumption and progress; procurement supplies commitments; finance supplies postings/payments. Analyse by company, project, location, work item/WBS, resource type and period. Map client BOQ lines to internal work items with explicit, versioned allocations; they are not necessarily one-to-one.

Simple UI: select an item and location once, then see Budget / Incurred / Paid / Forecast with expandable resource and source detail. Never label all four as Actual. Pages 30-31 define the measures and a reconciliation example.

Cost dictionary: compare like with like

Proposed definitions for agreement with cost control and finance. These are requirements, not a claim that every measure is implemented.

Measure	Meaning and control
Original budget	Frozen approved cost plan: quantities x resource rates, plus explicitly allocated indirect cost and risk. Keep the original version.
Revised budget	Original budget plus approved budget changes/transfers. An owner selling-price variation must not automatically become an equal cost-budget change.
Plan to date / earned value	Planned value is budgeted work scheduled by the status date. Earned value is budgeted value of accepted work completed. Both must use the same approved baseline and measurement rules.
Commitments	Approved purchase/subcontract obligations, including controlled amendments. Show total and remaining uninvoiced/unperformed exposure separately; total commitments are not extra cost to add to actuals.
Incurred actual cost	Posted cost plus approved period-end accruals, net of corrections and credits, counted once. Stock receipt, issue, installation and invoice may represent different stages, not four costs.
Cash spent / received	Allocated payment or receipt amounts and dates. Separate advances, taxes and retention; show reconciliation status. Cash spent may precede or follow the related incurred cost.
Cost to complete	Reviewed remaining committed work + remaining uncommitted work + explicit risk allowance. Identify assumptions, rates, productivity, owner and update date.
Forecast final cost / EAC	Incurred cost + cost to complete. Show variance to revised budget; a favourable forecast is not realised profit. Keep prior approved forecasts.
Revenue / margin	Separate contract value, approved/pending changes, claimed, certified, invoiced and collected amounts. Compare compatible revenue/cost forecasts for margin; cash balance is a separate measure.

Every number needs a basis

Display cutoff date, company/currency, tax basis, unit of measure, baseline/revision, allocation status and data freshness. For each cost type expose budget quantity/rate, actual quantity/rate, productivity, waste and the source documents. Use material, labour, plant, subcontract and indirect-cost categories.

Never hide missing allocations, stale forecasts or unreconciled differences. Shared costs require an approved allocation basis and version history. Accruals must reverse or clear when the underlying bill arrives. A business owner must approve the precise posting and valuation rules.

One item, four different answers

Illustrative acceptance fixture only - invented figures, not extracted from JAKH. Example: one floor slab work item, one cutoff date, AED, tax excluded.

Question	Example answer / explanation
What was planned?	Original budget 100,000; approved cost-budget change +10,000; revised budget **110,000**.
What has it cost so far?	Posted cost 52,000 + approved unbilled accrual 8,000 = incurred cost **60,000**.
What is committed?	Approved package/order obligations total 80,000. Of these, 40,000 is already included in incurred cost; 40,000 remains. Other incurred labour/plant costs are 20,000.
What cash has left?	Payments **45,000**: 35,000 settling bills plus 10,000 advance. This does not turn incurred cost into 45,000 or 105,000.
What will remaining work cost?	Remaining committed work 40,000 + other remaining work 5,000 + risk 3,000 = cost to complete **48,000**. Future cash timing also reflects unpaid incurred costs and advance recovery.
Where will it finish?	Incurred 60,000 + remaining 48,000 = forecast final cost **108,000**. Revised budget 110,000 - forecast 108,000 = forecast headroom **2,000**.

Break the same item into resources

Expand the slab into concrete, reinforcement, formwork, labour, plant and allocated site overhead. Each has planned quantity/rate, actual consumed quantity, accepted installed quantity, loss/rework, incurred cost and remaining forecast. Use a common location and cutoff; aggregate the detail back to the item, floor and project.

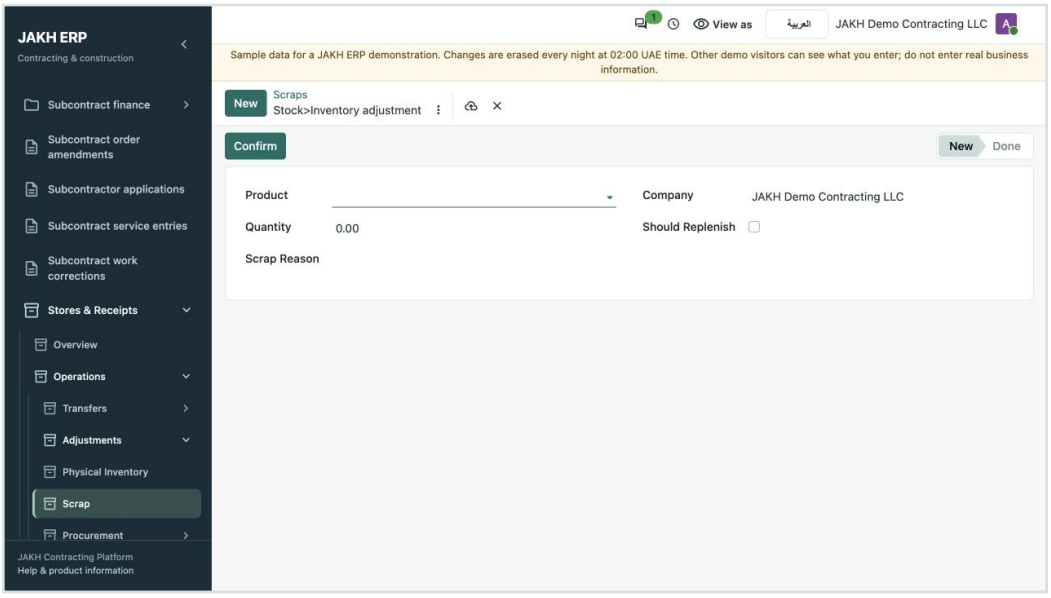
Required reconciliation tests

1. Every item total equals its contributing facts, with no source counted twice.
2. Cost posted without an item remains visibly unallocated until an authorised allocation.
3. A later bill replaces/clears the accrual rather than adding a duplicate cost.
4. Paid amounts trace to invoice/advance allocations and bank reconciliation.
5. Changes preserve the original budget and prior approved forecasts.

For quantity control, reconcile opening usable stock + receipts/transfers in against installed or otherwise consumed quantity, returns/transfers out, closing usable stock and recorded loss. Define measurement units and avoid counting waste again if it is already included in a consumption category.

6. Waste management

Existence: a stock-scrap function is visible; full construction waste control is not demonstrated. E77-E78.



E78. Unsaved scrap form: product, quantity/unit, reason, company and replenishment. It was discarded without entry or confirmation.

Two needs must be covered

Material efficiency: distinguish allowed waste, avoidable loss, damage, theft, rejected work, recoverable offcuts and rework. Link issued/returned/installed quantities to the item, location and responsible trade. An inventory adjustment alone does not explain what happened on site.

Site waste handling: classify waste, segregate/store it, arrange collection, record weight/volume, carrier and destination, then attach acceptance/weighbridge/disposal evidence. Track reuse, recycling, recovery proceeds and disposal cost. Confirm jurisdiction and contract requirements before specifying controlled waste paperwork.

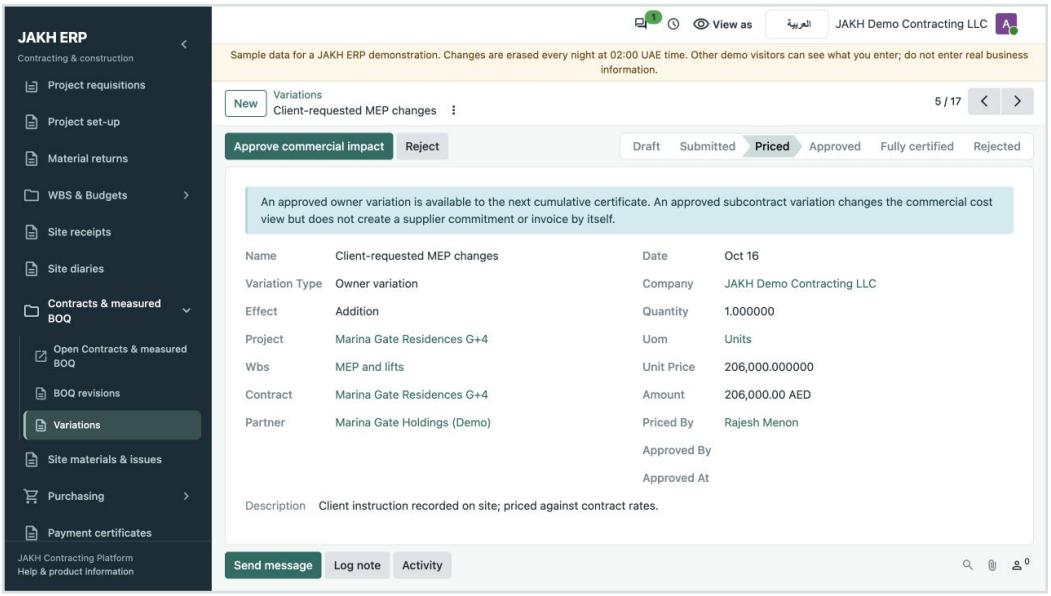
Needed workflow and simple UI

Site/store staff record waste against work or an NCR; the site/environmental owner selects the handling route; an authorised person approves the write-off or disposal; receipt/evidence closes the movement; cost control allocates the impact. One Project -> Site -> Waste register should connect physical movements and cost, with stock transactions remaining controlled linked records.

Test: reconcile reinforcement received, used, returned, reusable offcuts and lost/scrapped quantity for one floor; follow one disposal load to acceptance evidence and its cost. Define waste-rate denominators by material and units; do not combine tonnes and cubic metres in one total.

7. Variations and change requests

Existence: commercial variation records confirmed; the upstream change-event and time-impact chain remains partial. E85-E86.



E86. A priced owner variation links project, WBS and contract. The instruction is mentioned in free text; structured notice/time-impact fields are not visible.

What exists

The register includes Draft/Submitted/Priced/Approved/Fully certified and Rejected stages. The UI explains that an approved owner variation feeds certification, while a subcontract variation does not itself create a commitment or invoice. Preserve that distinction. Internal design changes, early warnings and potential changes were not demonstrated as a separate connected intake.

Needed workflow and simple UI

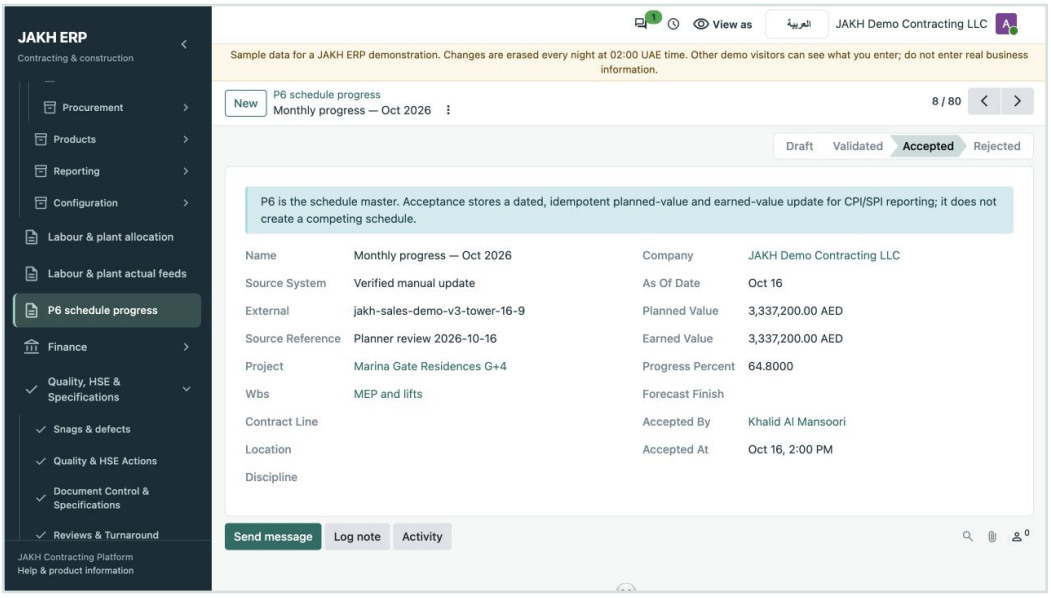
Essential. Capture a change event from a client instruction, RFI, revised drawing or site condition. Identify affected scope/location, initiator, responsibility and contract references. Track notice requirements, cost/time assessment, quotations, review, decision and implementation. Keep potential, instructed, submitted, approved, disputed, rejected and withdrawn outcomes understandable.

QS/PM own assessment; planning owns time effects; the designated internal/client authority owns decisions. Link approved outcomes to budget revision, contract/BOQ, subcontract/PO amendment, programme revision, forecasts and certificates. Internal approval and client acceptance are separate decisions. Configure notice dates from the actual contract; do not hard-code a universal deadline.

Test: an urgent instructed change can proceed under recorded authority while its price is disputed; costs, notice dates and programme effects remain visible. Use one Changes workspace with an evidence timeline, not repeated data entry in each downstream record.

8. Programme & delay

Existence: progress ingestion and acceptance confirmed; full planning and delay management not demonstrated. E79-E80.



E80. The screen says P6 is the schedule master. This accepted manual update has PV/EV and progress; forecast finish and location are blank. No activity-level link is visible.

Separate progress reporting from controlling time

A percentage or SPI does not explain the critical activity, lost days or recovery plan. The inspected record is a dated WBS progress fact. It does not demonstrate dependencies, calendars, baseline comparison, critical path, float, resource loading, look-ahead constraints or an EOT assessment.

Needed workflow and simple UI

Essential. Planner owns an approved baseline and dated updates. Site teams report actual starts/finishes and remaining work, linked to accepted evidence. Surface a near-term look-ahead with prerequisites such as design release, access, materials, inspections, labour and plant. Record delay events with discovery/date range, affected activities, cause, evidence, responsibility, mitigation and review status.

Connect delay assessments to notices, change events, the programme version analysed, forecast finish and any time/cost claim. Keep requested, assessed and granted extension distinct. A planner/contract administrator must validate analysis and contractual interpretation; a metric alone must not decide entitlement.

Architecture decision: retain a clearly defined P6 exchange or build original scheduling capability, according to operational need. Either requires one authoritative programme and traceable versions. **Test:** a late approved drawing blocks a floor activity; show the dependency, mitigation, notice and updated finish without overwriting the baseline.

9. KPI management

Existence: metrics and dashboards confirmed; a complete KPI definition/ownership/action cycle is unverified. E53-E54, E66, E82-E84.

Construction dashboards already expose CPI/SPI and financial/physical progress. Saved Project dashboards emphasise tasks, logged hours and task closure, while document turnaround has reviewer/due-date data. These are useful components, but a KPI needs an agreed meaning and accountable response.

Recommended starting KPI	Definition / owner / response
Cost performance	CPI = earned value / compatible incurred actual cost, where cost > 0. Cost controller investigates adverse cost/productivity drivers; show excluded/unallocated costs.
Schedule performance	SPI = earned value / planned value, where plan > 0; planner also tracks milestone slippage and critical constraints. SPI is not a count of delay days.
Completion forecast	Revised budget minus forecast final cost, plus forecast coverage/freshness. QS owns the cost-to-complete review and corrective action.
Quality	Open/overdue NCR count, closure duration, recurrence and rework cost. QA/QC owns review; normalise comparisons by work/inspection volume where appropriate.
Buying / subcontract	Required-on-site delivery reliability and overdue package actions. Procurement/PM owns expediting and recovery.
Cash / commercial	Overdue receivables/payables, certified-but-unbilled value and forecast cash shortfall. Finance/QS own collections, billing and payment planning.
Resources / waste	Productive asset hours / available planned hours; material waste / defined input quantity. Plant/site owners investigate downtime and loss.
Approvals / records	Overdue reviews and turnaround by document type/owner; current approved document coverage. Document control and workflow owners resolve bottlenecks.

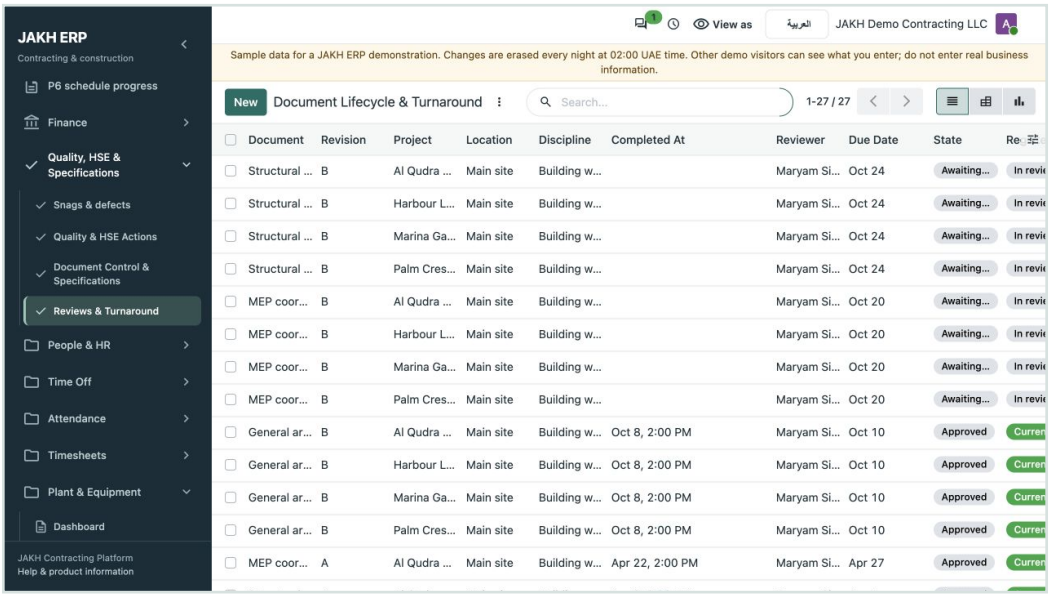
One metric definition, several contextual views

Store formula/version, unit, population, exclusions, period, baseline, target, threshold, owner, refresh time and source link. Support trend and explanation, then assign an action with due date. Start each role with a few actionable measures; let users expand the supporting detail.

Test: a dashboard total reconciles to its records under the same filters; zero, missing and not-applicable values differ. Crossing a threshold creates one owned action and stops duplicate reminders. Targets need business agreement; this study does not invent performance benchmarks.

10. Keeping records

Existence: document revisions, attachments, site diaries and review records confirmed; retention and distribution controls remain unverified. E37-E45, E57-E59, E82.



E82. Review records expose document/revision, project, reviewer, due date, completion and status; age/cycle measures are available in the wider table.

The record must explain what was known then

Preserve a document ID and revision, issue purpose, creator/source, project/location/package, timestamps, review decision and linked work. The current approved drawing should be clear alongside history. Approval and issue-for-construction are separate meanings and must be labelled consistently; the register exposes both fields.

Needed workflow and simple UI

Foundation across all modules. Capture -> classify -> review/revise -> approve -> issue/distribute -> acknowledge where required -> supersede/archive. Link correspondence, photos, inspections/tests, daily reports, instructions, notices, measurements, meetings, commercial evidence and handover packs. A chatter attachment is not by itself a controlled document register.

Document control owns numbering/revision/distribution; originating teams own content; project/finance owners set retention and access rules. Search by project, floor, work item, party, date and record type. Define offline photo capture/sync, permissions, version conflicts, backups, export and retention/hold behaviour in engineering validation.

Test: reconstruct which approved revision and instruction supported a particular pour/certificate on a historical date, retrieve its evidence, and export an indexed package without losing relationships. Show relevant records inside each workflow; keep one authoritative copy.

11. Approval workflow

Existence: several document-specific routes and approver histories are visible; universal queue coverage and enforcement remain untested. E15, E54, E59, E71, E82, E90.

Foundation. Use one approval service and personal queue, with distinct policies for each business object. The table below proposes ownership for validation; it is not a statement of the current permission configuration.

Object / trigger	Proposed responsibility and decision boundary
Tender / budget / forecast	Estimator or QS prepares; PM/commercial authority reviews; delegated authority approves within value/risk limits. Preserve priced and approved versions.
Purchase / subcontract award	Requester -> technical/budget check -> procurement/commercial -> authorised award. Receiving and paying require separate checks.
Inspection / NCR / snag	Site/contractor submits -> QA/QC/technical review -> required consultant/client sign-off -> verified closure. Configure independence and hold-point rules.
Change / time assessment	Originator -> QS and planner assessment -> PM/internal authority -> client/consultant decision where needed. Keep commercial and time decisions traceable.
Measurement / certificate	Site validates quantity -> QS assesses -> commercial/finance checks -> authorised certificate. Supplier claim, certificate, bill and payment remain distinct.
Bill / payment / release	Match supporting obligation and evidence -> finance review -> authorised payment/release -> bank reconciliation. No dashboard permission grants bank authority.
Document / handover	Author -> discipline reviewers -> approval -> authorised release. Handover also checks technical, client acceptance and commercial obligations.

Controls and UI that every route shares

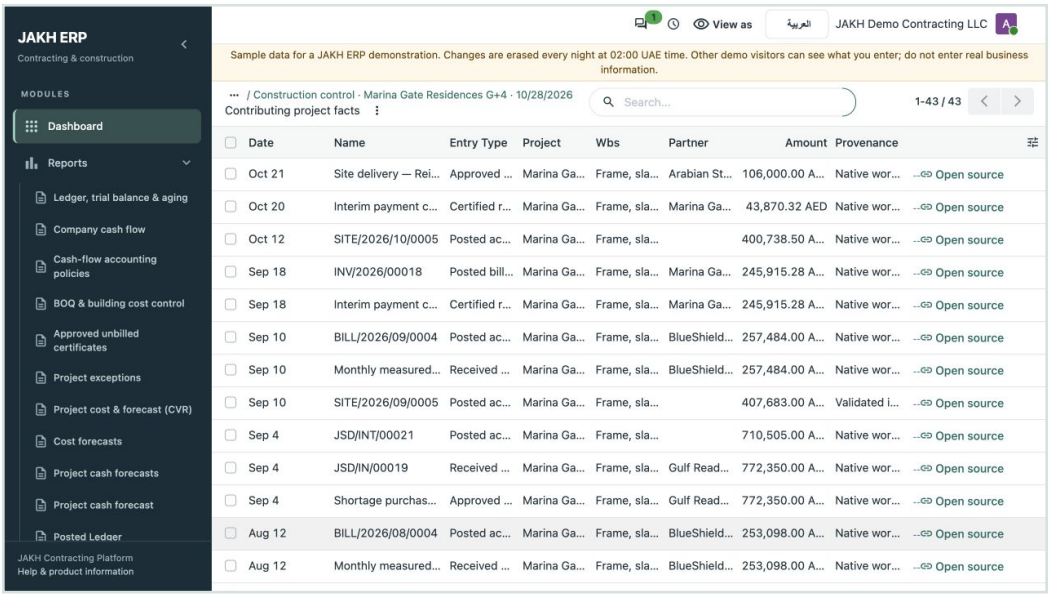
Show current step, assigned person/role, due date, blockers and decision history on the record and in My Work. Support sequential/parallel review, amount/risk limits, substitutes, absence delegation, escalation, rejection reasons and resubmission. A material edit must invalidate or reroute the affected approval version.

Record policy version, actor, authority, timestamp, comments and document snapshot. Enforce permissions on the server and prevent duplicate/concurrent decisions. Distinguish an operational activity marked Done from a formal business approval.

Test: absence delegation, out-of-limit approval, prohibited self-approval, changed documents and simultaneous clicks all produce correct, explainable outcomes. Queue counts reconcile to pending business records. Read-only inspection did not execute these tests.

12. Tracking and transparency

Existence: source drill-through and provenance confirmed; complete event history and permission boundaries require validation. E69-E72, E81-E82, E90-E92.



E69. Contributing facts show dates, classifications, project/WBS, amounts, provenance and Open source links. One was followed to a supplier bill and certificate.

Keep the good traceability, make it easier to read

The project-control audit lists financial/control facts, including their provenance. This is useful evidence lineage; it is not proof of an immutable who-changed-what audit log. Approver/date fields and chatter support some history, but universal old/new values, deletions, permission changes and restoration were not tested.

Needed behaviour and simple UI

Foundation. Every record answers: what is this, which work does it affect, what changed, who owns the next step, when is it due, and why is it blocked? Use a consistent timeline for submission, revision, decision, execution, posting and settlement. Link each event to its actor, authority, source version and resulting records.

A project overview should summarise exceptions and next actions; users expand only what needs explanation. Every aggregate drills to the same filtered records and shows refresh/cutoff and missing data. Client/subcontractor views show their permitted scope, with confidential internal prices and personal information controlled by policy.

Test: trace a forecast variance to an item, bill/service entry and originating scope change; reconstruct its approval version. An authorised user can export that chain, and an unauthorised user cannot gain access through a deep link. Records/approvals/tracking share one history service rather than three duplicate modules.

Define an original application

Rebuild boundary and migration study. The current product supplies observable business requirements, not implementation code.

No Odoo application-code dependency

Build original source, data models, APIs, approval/state rules and UI. Do not copy, translate or adapt Odoo modules or application code. No existing source code was inspected during this study. Selection of framework, database and deployment architecture remains a later engineering decision.

Core domain relationships to specify

Company and authority -> party/person -> opportunity/tender -> estimate revision -> contract and BOQ baseline -> project, location and WBS/package -> demand and commitment -> delivery/service/progress evidence -> variation/certificate -> invoice/payment -> handover/defect obligation.

Keep document revisions, financial postings, approval events and operational evidence linked but distinct. Define stable identifiers, permissions, immutable approved snapshots, corrections, search, attachments and audit history across those relationships.

Explicit domain scope also includes quality/NCR, asset custody/maintenance/hire, waste, changes/notices, programme/delay, item/resource allocations, cash forecasts and KPIs. Company ownership, currency/rate snapshots, intercompany links, scoped report runs and verified findings are foundations. AI uses these controlled records and services; see pages 44-53.

Finance and supporting ERP scope

Specify original accounting/posting and reconciliation requirements with the finance owner. Inventory the required HR, attendance, leave, timesheet, plant, fleet, master-data and administration functions. Integration decisions require explicit scope; they must not quietly replace requested functionality.

Migration and cutover requirements

Step	Evidence required before production use
Inventory and export	Authorised exports of records, identifiers, attachments, companies, currencies, statuses, approvals and ownership.
Map and trial-migrate	A field/state map and repeatable trial import; preserve relationships and approved historical meaning.
Reconcile	Counts, opening balances, inventory, open orders, project budgets/costs, certificates, claims, retention and attachments checked by owners.
Pilot and cut over	One representative project plus exceptions; parallel reconciliation, backups, rollback and agreed transition responsibilities.

Validate before detailed design

The read-only study is ready for review. Detailed design and application coding remain behind the requested review point.

Priority sequence - the full scope remains

Priority	Outcome
First: navigation and context	Agree roles, canonical records, project/floor/package context, terminology and an understandable work queue.
Next: tower workflow depth	Resolve engineering-to-site, instruction-to-variation/time, package-to-measurement, and technical-to-commercial closeout handoffs.
Then: coherent original ERP	Design and build the lifecycle in increments, including original finance controls and the required supporting modules.
Before replacement	Trial migration, representative-user validation, accounting/control review, security/access testing, integration checks and a controlled pilot.

Tasks for the next validation round

1. Find the approved drawing for one tower floor and confirm which revision may be used.
2. Request a material, follow a partial/rejected delivery, and find the installed quantity.
3. Trace a subcontract package from measured work through payment.
4. Record an instruction, identify cost/time consequences, and follow its approval or dispute.
5. Explain claimed, certified, invoiced and collected values for one period.
6. Find all snags in one unit, verify the fix, and check partial-handover readiness.

Use you and representative estimator, site, procurement, QS/PM, quality/document-control and finance users where available. Measure unassisted completion, time to the correct information, wrong turns, repeated entry and missed handoffs. Use the proposed targets on page 13, record a real baseline, and agree gates before testing; no staff usability test has yet been run.

Decisions still needed

Confirm actual roles/approval authority, contract types, floor/unit structure, consultant/client/subcontractor participation, document and programme tools, accounting/localisation needs, languages/mobile/offline needs, project volumes and migration depth. A delivery estimate should follow this scope, not precede it.

Terms used in the study

BOQ: bill of quantities. WBS: work breakdown structure. QS: quantity surveyor. RFI: request for information. ITP: inspection/test plan. EOT: extension of time. DLP: defect-liability period. O&M: operation and maintenance.

Twelve validation scenarios

Proposed acceptance work after study review. These scenarios have not been executed against the current ERP.

Area	Evidence needed to accept the workflow
1. NCR	A failed inspection reaches authorised repair and verified closure; cause, effectiveness, location and rework costs remain linked.
2. Assets	One owned and one hired asset can be assigned, moved, maintained and costed; one installed asset has an exportable handover pack.
3. Cash	Partial payments, advance recovery, retention and late receipts reconcile between project forecasts, ledgers and bank evidence.
4. Subcontracting	A partly rejected claim produces the right certificate, deduction, bill and payment; package totals and final retention reconcile.
5. Item cost	One floor item reconciles original/revised budget, resources, incurred cost/accruals, remaining obligations, cash and forecast without duplication.
6. Waste	One material quantity balance closes; a disposal/recovery movement has ownership, destination evidence and allocated cost.
7. Changes	One instructed but disputed change retains notices, assessment, decision versions and downstream budget/order/programme links.
8. Programme / delay	One delayed prerequisite is traced to affected activities, mitigation, forecast finish and a versioned time assessment.
9. KPIs	A metric reconciles to its source population and produces one accountable action when its agreed threshold is crossed.
10. Records	A historical work event can be reconstructed using the correct document revisions, correspondence, photos and approval evidence.
11. Approvals	Test delegation, limits, rejection, changed documents and concurrency; the personal queue matches pending business decisions.
12. Transparency	A permitted user traces a dashboard total to source events; a user outside that scope cannot access it through links or exports.

Sequence and release priorities

First: agree IDs, location/work-item structure, definitions, records, approvals and history. **Then:** validate integrated project scenarios across all twelve areas; deepen NCR/waste, programme/delay and cash/item allocation where evidence is weakest. **Before release:** prove roles, reconciliation, migrations and recovery with representative users.

All twelve remain in scope. Advanced analytics and automation can follow reliable operational capture; essential site, cash, quality and approval controls must not be postponed merely for a cleaner-looking first screen.

Evidence and limits

Primary source: live UI at https://demo.jakh.net/ on 28 October 2026. Evidence IDs match saved screenshot and accessibility-snapshot prefixes.

Evidence	Observed route / role / purpose
E02, E39, E41	Administrator dashboards and complete expanded sidebar. E41 is an accessibility inventory, not a single screenshot.
E05-E07	My Company / USD: estimate EST/2026/00004 -> linked project and project shortcuts.
E15-E17	My Company / USD: measured client certificate -> contract -> cash/closeout tab.
E19-E22	My Company / USD: material-shortage requisition -> P00005 -> JD2/IN/00001 receipt.
E28-E29	My Company / USD: subcontract service entries and submitted review record.
E30-E38	Site Engineer, JAKH Demo Contracting LLC / AED: project portfolio, G+4 project, snags and diary.
E40, E42	Administrator, fuller company: G+4 construction report and schedule-progress register.
E43-E45	Administrator, fuller company: document register, G+4 drawing revision history and category menu.
E46-E49	Administrator, fuller company: variations and the approved completed-project final account.
E50-E52	Estimator: role dashboard, CRM entry and eight BOQ studies.
E53-E54	Project Manager: role dashboard and two-item activities/approvals queue.
E55	Administrator returned to original My Company context and normal browser viewport.

Scope limits that affect the conclusions

The first administrator view selected My Company (USD); role previews opened JAKH Demo Contracting LLC (AED), which has fuller samples. Empty initial lists are not proof of absent features. The study inspected a G+4 building and villa/fit-out examples, not high-rise operating scale.

Only navigation, view filters, role previews, record/form reading and report generation/viewing were used. Unsaved scrap and report criteria were discarded. Generated report-view records are the only new records from these inspections. No business document was edited, approved, posted, paid, closed or deleted; no source-code audit, migration, reconciliation or state-transition test was performed.

Accessibility observations are risks to verify: truncation, small/dense text, colour-dependent cues and tab-state semantics. Keyboard/focus, screen-reader behaviour, measured contrast and error recovery were not tested. Screenshot blank capture margins are omitted only in PDF placement; visible content is unchanged. Early clipped capture E07 is excluded from visual conclusions in favour of E33.

Follow-up evidence and reference

Administrator in JAKH Demo Contracting LLC / AED, then restored to My Company. Public reference reviewed separately; no ERP data was sent to it.

Evidence	Observed purpose / result
E56-E59	Quality register -> building Non-conformance -> Resolution -> Lifecycle. Populated, closed case; no transition executed.
E60	Public QIC NCR toolkit product description, accessed 28 October 2026. Paid documents not inspected.
E61-E64	Equipment register -> tower crane -> Product Information -> Maintenance. Identity, cost, location gap and reliability fields.
E65-E69	BOQ report selector -> dashboard-generated building report -> detail/optional columns -> source facts. Selector did not offer the existing project; dashboard route succeeded.
E70-E72	Source fact -> subcontract bill -> supplier certificate -> finance agreement. Quantities, deductions, approvals and balances.
E73-E76	Company cash report -> policy prerequisite error -> empty policy register -> empty project cash-forecast register.
E77-E78	Empty scrap register -> unsaved form, then discard. No scrap or inventory adjustment created.
E79-E80	Schedule register -> accepted building progress record. P6 named as master; manual source shown.
E81-E84	Project-control facts, document turnaround and saved generic dashboards. Record lineage differs from full change-event auditing.
E85-E86	Variation register -> priced building MEP change. Commercial lifecycle and scope links observed.
E87-E90	WBS -> budget record -> forecast register -> approved structural forecast. Authoritative dated forecast differs from simple WBS field.
E91-E93	Matched receipt sources -> posted journal entry -> original dashboard/company restored. No payment/reversal action executed.

Source and interpretation limits

QIC Management Systems: NCR Management Toolkit public description. Its form/procedure/register/dashboard outline informed the NCR comparison. The requirements and UI placement in this report are our proposed construction brief; they do not reproduce or validate the paid toolkit or establish ISO compliance.

An empty list applies only to the inspected role/company/filter context. Fields or capabilities not visible in sampled records are labelled unverified, not globally absent. Sample totals and same-person approvals are evidence of these examples, not proof of accounting correctness or permission enforcement. The public demo can change or reset. Detailed design and coding remain subject to your study review.

Multiple companies, one clear context

Revision 4 requirements. Earlier company switching was observed; entity isolation, intercompany accounting and consolidated reporting were not tested.

Manage several legal companies in one ERP without mixing their obligations. Keep the four starting points. A persistent company selector scopes the workspace; the project and every transaction show their owning company. My Work can collect authorised tasks across companies, with a company label on every row.

Requirement	Proposed behaviour
Separate legal entities	Each company has its own ledger, accounting currency, fiscal periods, tax/localisation setup, numbering, bank accounts, approval authority and access scope. A branch/site is not automatically a separate legal company.
Project ownership	Default each project to one legal company. A shared project or joint venture uses a management umbrella with explicit company-specific contracts, costs and allocations; an umbrella does not merge legal ledgers.
Transaction ownership	A purchase request, PO, bill, payment, stock movement or employment record has explicit legal ownership. The project normally supplies it. Resolve a conflicting company selection before saving; never silently move a record between companies.
Shared vendor identity	A vendor may have one shared party identity, with company-specific approval status, agreements, currencies, payment terms, tax setup and permitted payment instructions. Restricted balances/contracts remain scoped.
Central buying / shared resources	One sourcing exercise can serve several companies, but awards, orders and allocations remain entity-specific. Cross-company purchases, stock, labour and plant charges use linked intercompany documents and reconciliation, not informal transfers.
Access and group visibility	Roles include company + project + duty. Group management access is explicit. Enforce scope in records, reports, search, exports, attachments, AI retrieval and caches; a selector is not the security boundary.
Company switching	Switch the workspace context without changing an open document's owner. Handle unsaved edits explicitly; label deep-linked records with their actual company. Group scope is for authorised overview/reporting, not ambiguous transaction entry.

Simple experience: Company A / Tower 01 stays visible while the user requests material or reviews a claim. Company B appears only after a deliberate switch or an explicitly labelled cross-company queue. Company-wide views include central costs and unallocated work as well as projects.

Vendor currency follows the purchase

Purchase-request design: vendor information provides defaults; the accepted quotation or agreement supplies the transaction currency.

Step	What the user sees and what the system does
1. Choose project / company	Use the project's legal company and budget context. Allow a documented central-buying/intercompany route where required, rather than assigning another company's bill to the project silently.
2. Capture the need	Enter item/service, quantity, unit, required date and work item/location. A vendor is optional at requisition stage; without one, show an estimate in the company budget currency.
3. Choose vendor / quotation	Currency precedence: selected valid quotation -> applicable vendor agreement -> vendor preferred permitted currency for this company. If unresolved, ask for currency before submission. Never infer it just from vendor country.
4. Show both values	Show vendor-currency price/total and the company-currency estimate. Put rate direction, date/source and provisional basis in clearly labelled rate details; surface missing/stale-rate warnings directly. Taxes, freight, duty and other charges are explicit, not hidden inside FX.
5. Compare and approve	Compare offers in a common approved evaluation currency/date with equivalent quantities, scope, delivery and landed-cost assumptions. Budget/authority checks use the company's policy basis. Changing vendor, currency, price or FX beyond policy triggers a visible recheck and, where required, renewed approval.
6. Create the order	One PO has one purchasing company, vendor and transaction currency. Split incompatible selections into linked orders. Carry the chosen quotation, currency, terms, quantities and approval version forward without retyping.

Defaults should save effort without hiding decisions

A vendor may quote in several currencies. Store allowed currencies and date-effective terms; a EUR quotation can override a USD preference when valid. Changing the vendor must not reinterpret an old USD amount as EUR. Reprice or request confirmation with a clear before/after comparison.

One requisition can collect needs before sourcing; quotations retain their own currencies. Multiple projects within one company need explicit allocation. Multiple companies require separate authorised purchasing obligations. Bill/payment currencies and bank details are validated independently; a currency default does not authorise a bank-detail change.

Currency: keep the four meanings clear

Illustrative acceptance fixture only. All rates below are invented, not current exchange rates or JAKH transaction data. Taxes, fees, accruals and prior revaluation are excluded.

Company A uses AED for this example; the vendor quotes EUR 40,000. The selected quote controls the order currency. Exchange-rate snapshots explain why the company-currency values can change through the lifecycle.

Stage	Rate: AED per EUR	Meaning
Request estimate	4.00	EUR 40,000 x 4.00 = AED 160,000 for the request's budget check. This is neither a posted cost nor a payment.
Order approval	4.10	AED 164,000 commitment equivalent. Against an AED 170,000 item budget, remaining headroom is AED 6,000. Recheck the approval policy.
Bill recognition	4.15	Illustrative liability carrying amount AED 166,000. Posting/valuation and cost allocation follow the approved accounting policy.
Settlement	4.20	AED 168,000 cash payment. With no intervening revaluation/fees, the illustrative settlement difference is AED 2,000; do not rewrite the original request or bill rate.

Four currency roles, shown only when relevant

Transaction currency: what the quotation, order or invoice is denominated in.
Company accounting/functional currency: the company's approved accounting basis.
Project control currency: an optional agreed comparison basis for budgets and forecasts.
Report presentation currency: the selected reporting basis, with its translation policy and date.

Required controls

Store original amounts and currency alongside rate, direction, source, effective date and converted amounts. Use decimal precision and currency-specific rounding. Keep budget, commitment, recognition, settlement and reporting rate snapshots distinct. Reports separate operational variance from FX effects.

Missing/stale rates must produce an explicit exception before an amount-dependent approval or posting; a report must not silently treat them as 1:1 or zero. Finance controls rate overrides, remeasurement, realised/unrealised FX treatment and period locks. Historic approved reports remain reproducible.

Accounting reference: IFRS Foundation: IAS 21 distinguishes foreign transactions, functional currency and presentation currency. Applicable standards, jurisdictions and posting policies must be agreed with the finance owner; this example does not establish compliance.

One reporting system, several scopes

Project and company reporting are core requirements. Authorised group reporting is an additional explicit scope, with consolidation controls.

Start a report from the relevant project area or Company > Reports. Use the same definition and metric rules. The header always shows company scope, project scope, period/as-of date, currency basis, document states, filters, data freshness and whether the population is restricted by permissions.

Scope	Examples and required behaviour
One project	Budget versus incurred/paid/forecast; buying and subcontract status; cash outlook; programme/delays; NCRs, waste, assets, approvals and handover readiness. Filter by item, package, floor/system, party and period.
One company	All its projects plus non-project activity, overheads and unallocated amounts. Finance, procurement, people, stores and asset reports use the same source facts and respect restricted personal/financial detail.
Selected projects	Portfolio comparisons within authorised companies. Show included projects and exclusions; expose incompatible baselines, cutoffs or missing forecasts before presenting rankings.
Selected companies / group	Translate to an explicit presentation basis. Label a management aggregation separately from a consolidated financial report. Consolidation requires defined scope, mappings, aligned periods/policies and intercompany eliminations; adding company totals is insufficient.
Source drill-down	Summary finding -> contributing rows -> source record/version. Preserve company, project, cutoff and filters on every step. A reader with narrower rights receives an explicitly scoped report, not hidden-data snippets.
Saved / exported reports	Save a versioned report run with parameters, data cutoff and calculation/rate versions. PDF and spreadsheet exports include the same findings summary and scope; exports and shared links enforce access and distribution policy.

Aggregation must retain meaning

Never add unlike currencies or units. Recompute ratios from compatible numerators/denominators rather than averaging project percentages. Show unallocated and stale data. Reconcile project totals to company totals through non-project costs, allocations and adjustments; explain intercompany eliminations in group totals.

Changing filters or currency creates a matching summary for the new result. An asynchronous summary must never stay above a different report version. Where a report genuinely has no project dimension, say so and offer the relevant company report.

Every report starts with findings

A standard Report findings section sits above the details in every report, including exports. It works even when generative AI is disabled or unavailable.

At the top	Content contract
Scope and reliability	Company/project, reporting period, currency/tax basis, comparison basis, data freshness and coverage. Put material missing-data or reconciliation warnings before reassuring conclusions.
Main findings	Three to five concise findings ranked by business impact: what changed, size of variance, affected work and source. More critical exceptions remain visible through an explicit count/list.
Evidence and interpretation	Each factual statement links to the supporting metric/rows. Separate observed facts, forecast assumptions and suggested actions. A correlation or large variance is not automatically a proven cause.
Action / ownership	Name the accountable role or actual assigned owner when known, the relevant due date and a link to investigate. Suggestions remain suggestions until the user creates or assigns work.
Empty / limited results	State No matching records, Missing baseline, Incomplete data or Summary unavailable accurately. Never present no data as zero cost, good performance or completed work.

The logic reads the complete report result

1. Resolve authorised scope and filters; create a consistent report snapshot.
2. Calculate metrics across the full result set, not only the visible page.
3. Apply versioned report-specific rules for variances, trends, overdue items, concentration, missing data and reconciliation.
4. Build structured findings with values, units, comparison bases, severity and evidence references.
5. Render a plain-language summary; optional AI can explain the verified findings.
6. Check every generated numeric/factual claim against the findings before display. Fall back to deterministic text when validation fails.

A report definition includes its summary rules

Cost reports compare budgets and completion forecasts; cash reports explain dated inflows/outflows; quality reports surface severity, overdue cases and recurrence; programme reports show milestones/constraints. Define meaningful rules for each report instead of applying one generic prompt to every table.

Store report-run ID, metric/rule versions and, when used, model/prompt versions. Cache by exact report snapshot, scope and permissions. Do not let free-text records instruct the summariser to change filters, totals or its evidence requirements.

Examples of useful report summaries

Invented examples to specify behaviour. These are not findings from the live JAKH demo.

Project cost and cash report

Company A / Tower 01 | 30 September | AED | tax excluded
Revised budget 1,000,000; incurred 620,000; cash spent 540,000; remaining forecast cost 460,000; forecast final cost 1,080,000. Forecast next-period receipts 120,000 and outflows 180,000.

What the user reads	Evidence / interpretation
Forecast final cost is AED 80,000 above budget (8%).	1,080,000 - 1,000,000 = 80,000; denominator is the revised budget. This is a forecast variance, not already incurred overspend.
Structure and MEP account for the forecast overrun.	Illustrative package variances: Structure +50,000; MEP +30,000; all others net zero. Link the package forecast versions; do not invent the underlying cause.
AED 30,000 of incurred cost still needs item allocation.	Included in the AED 620,000 project total, but item-level attribution is incomplete. Link the unallocated source facts and responsible cost-control queue.
Expected outflows exceed receipts by AED 60,000 next period.	180,000 - 120,000 = 60,000 before opening cash and financing. It is not the projected bank balance. Link the dated cash lines and assumptions.

Project quality report

Company A / Tower 01 | as of 30 September
20 open NCRs, including 6 overdue and 1 critical. All numbers refer to the same authorised project population.

Summary: Six of 20 open NCRs are overdue (30%). One critical case requires immediate attention by its assigned QA/QC owner. Four of the overdue cases relate to the same subcontract package; investigate its linked records before drawing a cause or performance conclusion. A comparable prior snapshot is unavailable, so no improving/worsening trend is claimed.

Links: Open overdue cases; Open critical case; View affected package. The summary identifies the case's owner and due date from the record. It does not declare work fit for release or close the NCR.

The summary must change when the evidence changes

Filter to another floor/company, revise a forecast, correct an FX rate or reduce the reader's access: recalculate the population and its findings together. A stale summary is replaced with a visible refresh state, never left above new totals.

AI where it helps construction work

Strategic goal: reduce searching and repeated entry, improve evidence completeness, and help people explain exceptions. AI stays inside the existing workflows.

Where / entry point	Useful assistance and required evidence	Human decision / value to measure
Tenders Review tender pack	Extract requirements, dates and exclusions from cited tender pages; compare revisions; draft a compliance checklist and estimate assumptions.	Estimator validates scope and commercial assumptions. Measure review time and missed obligations, not text volume.
Documents Find / compare information	Answer questions with document ID, revision, issue purpose and quoted location; compare drawings/specification text and draft RFIs from linked evidence.	Document/technical owner validates use and interpretation. Measure correct-document retrieval and review effort.
Site Draft daily report	Turn notes, voice or photos into a proposed diary with quantities, location, labour and plant; flag gaps and ask for missing facts.	Supervisor confirms extracted facts. Measure entry time, correction burden and complete location/work-item capture.
Quality / NCRs Review this case	Suggest classification and similar authorised cases; assemble inspection/specification evidence and draft corrective-action questions.	QA/QC owns cause, disposition and verification. AI cannot certify structural safety or close an NCR. Measure missing evidence and review time.
Plan / delays Explain the blocker	Summarise constraints, prerequisites and dated progress; highlight possible slippage and draft a delay chronology linked to programme versions.	Planner validates logic, mitigation and time assessment. Measure useful warnings and false alarms; predictions require sufficient history.
Changes Prepare assessment	Assemble instructions, RFIs, revisions and measured impacts; draft a change narrative and list missing cost/time evidence.	QS/PM and authorised parties decide value, notices and entitlement. AI cannot invent contract deadlines or approve a variation.
Handover Check this pack	Compare required deliverables with actual tests, snags, as-builts, O&M and warranty records by location/system.	Engineer/client confirms acceptance. Measure missing documents found and pack-preparation time; document presence alone is not technical acceptance.

Keep one clear business action per screen. Show assistance such as Find the drawing, Draft this report or Explain this difference only when relevant. Avoid a separate AI menu for every module and avoid a permanently competing chat panel.

AI in buying, money and company work

The same permission and company boundaries apply to every suggestion. Models explain and draft; controlled services calculate and validate.

Where	Strategic fit	Boundary / success measure
Buying / subcontracting	Extract quotation lines, currencies, exclusions and terms; compare like-for-like offers; draft package/application reviews from linked measurements.	Validate extraction and company-specific vendor terms. Normalised FX/landed-cost arithmetic is deterministic. Buyer/QS awards and certifies; measure review time and missed exceptions.
Budget & cost	Explain variances through linked quantities, rates, waste, allocations and forecast assumptions; propose clearly labelled scenarios.	Cost controller approves allocations and forecasts. Never invent reasons for a variance. Measure supported explanations and time to trace a total.
Finance / cash	Suggest invoice matching, possible duplicates and collection priorities; explain cash scenarios using due dates and observed payment history.	Finance verifies matches, adjustments and payments. Risk scores are signals, not fraud verdicts. Measure false positives, reconciliation effort and cash forecast error.
Assets & Fleet	Find assets by need/location; summarise maintenance history and expiry evidence; later test predictive servicing on sufficient failure/use history.	Plant owner controls fitness/availability and maintenance decisions. Measure downtime and useful alerts; begin with rules for known service dates.
Stores / waste	Suggest movement coding; flag unusual loss or consumption against accepted work; extract disposal evidence and reconcile missing documents.	Site/stores confirms movement, quantity and destination. Measure capture completeness, false alarms and avoidable loss.
People	Answer employee policy questions from current permitted policies; draft leave/time corrections and explain the employee's own records.	Keep personal data restricted. No autonomous hiring, discipline or worker scoring. Measure self-service completion and factual correctness.
Reports / KPIs / My Work	Explain verified report findings, summarise a case timeline and propose relevant follow-ups with sources and accountable owners.	Findings engine controls numbers; rules control priority and routing. Do not create duplicate tasks or send messages automatically. Measure faster decisions and accepted suggestions.

Cross-ERP search: one optional Ask JAKH entry can answer a scoped question across authorised records, with source links. It shows the active company/project and does not silently broaden them. Users can always complete the same core work through ordinary screens.

AI controls and rollout

Design requirements for the original ERP. No AI provider, model, contract or data-sharing configuration is selected by this study.

Control	Required behaviour
Authority	AI reads only the requesting user's permitted company/project/record scope. Drafts are separate from submitted records. Server-side workflow rules and approvals remain the authority for award, posting, payment, closure and external commitments.
Evidence	Answers link to source records/revisions and identify missing evidence. Forecasts and suggested explanations are labelled. An unsupported answer must say that evidence is insufficient; model confidence is not proof.
Data handling	Define company-specific AI permissions, approved providers/regions, retention and training-use restrictions. Minimise/redact payloads; do not share payroll, bank or commercially restricted material by default. Apply the same policy to prompts, logs, embeddings and exports.
Untrusted inputs / tools	Treat vendor quotes, uploaded documents and emails as data, never instructions to the assistant. Restrict tools to named operations; enforce scope independently. No arbitrary SQL, unrestricted browsing or blanket write access from a model.
Review and recovery	Preview any proposed change with sources before authorised acceptance. Revalidate permissions and current version at execution; prevent duplicate submissions. Log model/version, relevant evidence, suggestions and user decisions with appropriate retention.
Reliability / cost	Measure task correctness, source support, user correction burden, latency and cost per completed task. Version and regression-test changes; provide per-company disable controls. Core workflows and rule-based report summaries continue during AI outages.

Sequence by value and readiness

First: reliable company/currency data, metric definitions, report-findings rules, source-linked search and read-only case explanations.
Then: quotation/document extraction and reviewable drafts for diaries, NCRs, RFIs, claims and handover packs.
Later: predictive cash, delay, maintenance and anomaly signals, only after sufficient representative history and measured benefit.

Select pilots with the business owner; measure baseline effort and errors, then compare assisted work. Keep features that improve completion and correctness at an acceptable operating cost. AI coverage across the ERP is intentional, but every screen does not need a generated message.

Acceptance checks for the added scope

Proposed release gates, not tests already run. This extension specifies requirements; it does not claim these capabilities exist or are enforced in the current demo.

Scenario	Required result
Two companies / same vendor	Company-specific terms, bank authority, balances and permissions remain separate. A Company A-only user cannot retrieve B data through search, exports, summaries, AI or cached answers.
Shared project / central buying	The umbrella totals reconcile to entity-specific transactions. Intercompany documents match; group eliminations do not remove legitimate third-party costs or double-count internal charges.
No vendor / several currencies	An unsourced requisition stays an estimate. A valid quotation overrides preference explicitly; incompatible company/vendor/currency selections split into linked orders. No silent relabelling of amounts.
FX changes / missing rates	Reproduce request, PO, bill and payment snapshots in the worked example. Missing/stale rates cannot produce false totals or pass amount-based authority checks; changed terms recheck policy.
Project / company / group reports	Reconcile project + non-project + unallocated activity to company totals. Group translation and eliminations are explainable. Preserve scope/cutoff through drill-down and export.
Every report has a summary	The full filtered population drives findings, including pagination and restrictions. Values, units, denominators and evidence links agree with the report. Empty, incomplete, stale and zero are distinguishable.
AI error / malicious document	Test fabricated citations, wrong currency, hidden instructions, sensitive-data requests and attempted unauthorised actions. Reject unsupported output; return a safe deterministic summary or an explicit inability to answer.
Useful AI / service outage	Compare pilot task time, accuracy and correction burden against the baseline. On model failure, normal ERP work and calculated summaries still work; no phantom approval, duplicate task or silent external message.

Configuration still to establish: legal companies and jurisdictions, accounting/control/report currencies, shared-project model, group consolidation needs, vendor terms, FX sources and tolerances, report owners/thresholds, AI data policy and pilot users. These are implementation inputs; the requirements above are already in scope.

Sources: IFRS Foundation, IAS 21 overview informs currency distinctions. NIST, Generative AI Profile informs lifecycle evaluation/governance. OWASP, Excessive Agency supports limited permissions, functionality and autonomy. Accessed 28 October 2026. The detailed ERP mappings and tests are our proposed requirements, not standards certification.